

Government Reputation, FDI, and Profit Shifting*

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Abstract

Does a better institutional reputation allow a government to tax an expanding investment base more heavily or induce it to promise a lower tax? Lower contract risk is associated with lower taxes within countries and greater foreign investment across countries. To study how these patterns can arise together, I consider a government that announces a tax before firms invest but may revise it afterward. Firms cannot relocate installed capital but can shift reported profits once the tax is known. When taxes discourage investment, a better reputation makes a promised reduction more effective at attracting capital. I analytically show that this incentive makes the government announce a lower tax rate when its reputation is low. The ability to shift profits across borders limits the revenue that the government can gain by raising the tax after firms' investments while allowing firms to make substantial investments even under a poor government reputation.

Keywords: Commitment, Corporate taxation, Foreign direct investment, Government reputation, Multinational firm, Profit shifting

JEL codes: H21, H25, H26, F21, F23

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1 Introduction

A government with a reputation for honoring its tax promises can attract investment that a less credible government cannot. A larger investment base might appear to give it room to announce a higher tax. Yet a tax promise matters more to investors when it is more likely to be implemented. Credibility therefore changes the investment response to the announcement as well as the size of the base. Which effect governs the announced rate? This question is particularly relevant for multinational firms, whose productive investment may be sunk when taxes are chosen even though their reported profits remain mobile.

The data suggest that greater protection of investment need not be accompanied by a higher corporate tax. Within countries, lower risk of changes to investment contracts is associated with lower statutory rates. Across countries, lower average contract risk is associated with greater net FDI inflows relative to GDP, including with controls for income, population, and natural resource rents. Better protection therefore accompanies the investment that could enlarge the tax base, yet the tax relationship runs against the idea that a larger base necessarily calls for a higher rate. The tax and FDI estimates use different sources of variation and do not identify a common causal effect. They motivate the question of how greater credibility can attract capital while inducing the government to promise a lower tax.

Understanding this relationship requires separating productive investment from the profits available for taxation. In public country-by-country reporting data, jurisdictions with low corporate tax rates account for a larger share of multinational profits than of employment. This separation motivates treating investment and the location of reported profits as distinct firm choices. Once capital is installed, the ability to move reported profits can still limit what the government collects from a tax increase. The incentive to attract capital must therefore be analyzed together with the incentive to tax it after installation.

I study these incentives in a model in which the government announces a tax before firms invest but may revise it afterward. The government maximizes tax revenue and is either a commitment type, which must honor its announcement, or an opportunistic type, which can change the rate after investment. Firms do not observe the type. They allocate capital in anticipation of the implemented tax and choose where to report profits once that tax is known. Reputation is the probability they assign to the commitment type. A more credible announcement can attract more investment, but it also makes investment more responsive to the promised rate. The government's announcement must balance the revenue from a larger base against the investment it would lose by promising a higher tax.

I first show analytically that a small improvement from initially low reputation can lower the optimal announced tax. Consider an economy in which investors know the government is opportunistic but still invest because profit mobility keeps the tax below confiscation. The announcement initially has no effect on their investment. Giving a small probability to commitment makes the promised rate relevant to expected returns. When a higher tax reduces the marginal return to investment, a lower announcement can now attract capital. The government reduces its promise provided additional aggregate investment lowers the marginal return after accounting for its subsequent tax response. Because the announced and implemented

rates initially coincide, the investment base initially changes less than its responsiveness to the promise. The incentive to attract capital through a lower announcement therefore dominates the incentive to tax a larger base. Investment and the tax base rise as the announcement falls.

Profit mobility is central to this comparison because it determines the investment that can be sustained without commitment. Without shifting, the opportunistic type confiscates profits, and investors who know its type place no capital in the host country. Reputation then creates an investment base that would otherwise be absent. Under a condition on profit curvature, both investment and the optimal announcement rise with reputation. With shifting, the government can instead start with a positive investment base and a tax below confiscation, allowing the opposite announcement response. The analytical decline holds under small changes in parameters, and the numerical solutions extend it to a wider range of reputation in the benchmark. It also survives alternative treatments of shifting costs at common reference elasticities. Limits on the amount of profit firms can move can reverse the response, showing why the scope for adjustment after the tax is chosen matters beyond the response to a small tax change.

The one-period comparison takes the opportunistic type's incentive to revise as given. I then extend the model to two periods, allowing it to honor the first announcement to preserve future investment and revenue. Raising the implemented tax above the promise reveals its type, so later investors know that it can revise its next announcement. Honoring the promise preserves the revenue associated with a reputation for commitment. Profit mobility reduces the current gain from revision and can sustain substantial investment even when initial reputation is poor. Without shifting, a government may honor a high rate but attract little investment. The relevant comparison consequently includes the rate it promises, the probability that it honors the promise, and the investment it attracts. I determine these choices in equilibria where both government types make the same announcement, checking both their incentives to announce that rate and the opportunistic type's subsequent decision to honor or revise it.

Literature Review This paper contributes to three related strands of literature. First, it extends the analysis of time inconsistency in capital taxation by allowing reported profits to remain mobile after productive capital is sunk. [Chari et al. \(1988\)](#) explain why a tax policy chosen before investment need not remain optimal afterward. [Dovis and Kirpalani \(2021\)](#) study rules when private agents are uncertain whether a policymaker can commit and learn about its type from its actions. I use this distinction between a commitment type and an optimizing type to study corporate tax announcements. Firms' subsequent reporting choices affect both the tax the government would like to implement and the investment response to its promise. I characterize how these choices change the announcement that maximizes revenue, then determine whether the optimizing government honors that announcement to preserve future revenue. The contribution concerns the level and credibility of a tax promise under a revenue objective. It does not provide a welfare-maximizing Ramsey policy.

Second, the paper contributes to the literature on foreign investment under the risk of expropriation by distinguishing the mobility of reported profits from the mobility of productive capital. [Thomas and Worrall \(1994\)](#) and [Aguar and Amador \(2011\)](#) study how limited

enforcement constrains investment and capital accumulation. [Schnitzer \(1999\)](#) examines how the investor's control rights affect the scope for a self-enforcing relationship with the host government, while [Azzimonti \(2018\)](#) studies the political determinants of expropriation and investment. These papers explain why investors anticipate extraction and how future gains can restrain it. The reporting choice in my model gives firms an additional way to protect returns after capital is installed. This distinction is related to [Janeba \(2000\)](#), where excess capacity allows a firm to move production after governments set taxes. Here firms adjust reported profits while production remains fixed. [Langenmayr and Simmler \(2021\)](#) show how the entry of immobile firms changes the tax chosen for a common base. I study how beliefs about the implementation of a tax promise change the announcement at a given composition of firms. An extension allows firms to differ and establishes when the same tax response arises under a common rate.

Third, the paper adds government reputation to the analysis of international tax competition and profit shifting. [Peralta et al. \(2006\)](#) study the joint choice of corporate taxes and the control of profit shifting. [Devereux et al. \(2008\)](#) distinguish competition for capital from competition for reported profits, and [Johannessen \(2010\)](#) shows how these two margins shape asymmetric tax equilibria. [Mongrain et al. \(2023\)](#) explain how enforcement changes the elasticities of retained profits and capital, with opposing implications for tax rates. Holding the foreign rate fixed, I examine how reputation changes domestic taxing incentives even with a given shifting technology. A more credible promise changes the response of investment to the announced rate, while the ability to move reported profits changes the government's gain from subsequently revising it.

The closest comparison is [Mardan and Stimmelmayer \(2020\)](#), who show that the relationship between country risk and corporate tax rates can reverse with the cost of profit shifting. Risk in their model is an exogenous probability that an investment project succeeds, affecting expected profits and tax payments. In my model, uncertainty concerns whether the government will implement its announced tax. I derive conditions under which a small increase in the probability of commitment lowers the optimal announcement and characterize how implementation changes beliefs and future revenue. The contribution is to explain how an adjustable reported base affects the credibility of a tax promise and the investment that this promise can sustain. Broad correlations between country risk and tax rates alone cannot distinguish these mechanisms.

The treatment of reported profits is also informed by empirical work on multinational activity. The geographic concentration documented by [Garcia-Bernardo and Jansky \(2021\)](#) and [Garcia-Bernardo and Janský \(2024\)](#) motivates a comparison of profits and employment on identical reporting populations. Evidence on shifting costs, participation, and investment in [Kallen \(2023\)](#), [Bilicka et al. \(2024\)](#), [McClure \(2023\)](#), and [Wier and Erasmus \(2023\)](#) guides the numerical comparisons. Because these estimates concern different firm decisions, I distinguish the response of reported profits at fixed investment from responses in investment and participation when choosing the cost comparisons. This evidence disciplines the interpretation of profit mobility without treating a broad measure of contract risk as the probability that a tax promise will be honored.

Outline Section 2 presents the empirical facts, and Section 3 sets out the model. Section 4 analyzes the one-period and two-period economies, presenting each numerical solution with its theoretical results. Section 5 examines extensions and sensitivity to the assumptions. Section 6 concludes.

2 Empirical Facts

Three observations motivate the analysis. First, lower contract risk is associated with lower statutory corporate tax rates within countries. Second, countries with lower average risk receive more net FDI relative to GDP. Third, low-tax countries account for a larger share of multinational profits than of employment. The first two observations raise the question of why better investor protection might accompany a lower tax despite a larger investment base. The third explains why answering it requires distinguishing productive investment from reported profits. The tax and FDI comparisons use different sources of variation, whose interpretation I examine below.

2.1 Data and Measurement

I use the *Contract Viability/Expropriation* subcomponent of the International Country Risk Guide (ICRG), published by the PRS Group. Together with Profits Repatriation and Payment Delays, it forms the ICRG Investment Profile score. The subcomponent assesses the risk of unilateral modification or cancelation of investment contracts, including expropriation. Its original score ranges from zero, indicating very high risk, to four, indicating very low risk. Throughout the paper, *contract risk* denotes four minus this subcomponent score. It therefore ranges from zero to four, with higher values indicating greater risk.

The monthly source file covers May 2001 to December 2021. I form calendar-year averages and require all twelve months in the baseline. Complete annual scores, therefore, cover 2002–2021. Statutory corporate income tax rates, including subnational taxes, come from the Tax Foundation. Real GDP per capita, population, and natural resource rents come from the World Bank. Rates and rents as a share of GDP are measured in percentage points. The estimation sample requires these variables to be jointly observed and is held fixed when adding controls. Tax regressions do not require FDI or multinational profits to be observed. With risk lagged one calendar year, the annual sample contains 2,424 observations in 135 countries over 2003–2021. Appendix C.1 details the source reconstruction and country coverage.

Net FDI inflows relative to GDP come from the World Bank series `BX.KLT.DINV.WD.GD.ZS`. Net inflows include equity transactions, reinvested earnings, and intercompany debt. They can be negative and are not a direct measure of newly installed productive capital. I retain zero and negative flows. Matching FDI with lagged contract risk and the same three controls gives 2,552 country-year observations in 136 countries over 2003–2021. The FDI sample does not require a tax rate to be observed. Country averages use identical years for FDI, risk, and controls.

For reported profits, I use the OECD’s public country-by-country reporting aggregates, which cover large multinational groups, generally those with consolidated revenue of at least EUR 750 million. I compare profits and employment on identical foreign parent–host pairs

observed throughout 2017–2021, separately for profitable groups and all profit statuses. This comparison does not require allocating reported profits to hypothetical locations of production. Appendix C.5 documents the reporting populations, and Appendix C.6 examines coverage, tax thresholds, and country exclusions.

I also retain the 2016 estimates of net inward profit shifting from the misalignment model of Garcia-Bernardo and Jansky (2021). These estimates compare reported profits with profits allocated according to real economic activity. Appendix C.4 examines their relationship with contract risk and aggregate investment risk, defined as twelve minus the ICRG Investment Profile score. It reports the original specifications and additional controls on a common 116-country sample. These direct shifting estimates complement the observed geographic shares, but their relationship with risk is sensitive to controls and sample composition.

Interpreting the Measures. Contract risk concerns the treatment of investment contracts more broadly than corporate taxation. It is not a direct measure of the probability that a government will honor a particular tax promise. Similarly, a statutory rate observed after investment may already incorporate a policy revision, and an accounting effective tax rate can differ from the statutory rate without any broken promise. The comparisons below are descriptive relationships involving investor protection, observed taxation, and the location of profits. Appendix C.7 explains why identifying the effect of tax credibility requires dated announcements and subsequent implementation.

2.2 Fact 1. Higher Contract Risk Is Associated with Higher Tax Rates.

Within countries, a one-point increase in contract risk is associated with a statutory tax rate 0.781 percentage points higher, conditional on income and year effects. This positive relationship also survives each individual country exclusion. It runs against the simple argument that better protection, by attracting investment, should permit a higher corporate tax. The model asks how a more credible announcement changes the response of investment as well as the size of the tax base.

Let q_{ht} denote contract risk, y_{ht} real GDP per capita, and τ_{ht}^{stat} the statutory rate in country h and year t . I estimate

$$\tau_{ht}^{\text{stat}} = \alpha_h + \lambda_t + \beta q_{h,t-1} + \delta_y \log y_{ht} + \delta' Z_{ht} + u_{ht}, \quad (1)$$

where country effects α_h absorb persistent national differences and year effects λ_t absorb common changes. The additional controls Z_{ht} are log population and resource rents. Standard errors cluster by country. The lag aligns the score with the year preceding the observed tax rate. It does not make changes in risk exogenous to tax policy.

Table 1 adds the controls sequentially on the same sample. Column (1) controls for income. Columns (2) and (3) additionally account for country size and resource rents, which may affect the corporate tax base independently of contract protection. The coefficient declines from 0.781 to 0.641 and remains positive. The first two estimates are significant at the 5% level and the third at the 10% level.

TABLE 1: Contract risk and statutory corporate income tax rates

	(1)	(2)	(3)
Contract risk	0.781** (0.336)	0.664** (0.331)	0.641* (0.335)
Log income per capita	Yes	Yes	Yes
Log population	No	Yes	Yes
Resource rents	No	No	Yes
Country and year effects	Yes	Yes	Yes
Observations	2424	2424	2424
Countries	135	135	135

Notes: The dependent variable is the statutory rate in percentage points. Contract risk equals four minus the ICRG Contract Viability/Expropriation score and is lagged one year. All columns use the same 2003–2021 sample and include country and year effects. Country-clustered standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The coefficient remains positive over 2010–2019, among countries observed in every sample year, after excluding zero tax rates, and outside the original group of tax havens. Omitting each country in turn gives coefficients between 0.686 and 0.892 with income controls, all significant at the 5% level. These comparisons support the baseline sign, but separate country trends reduce precision. Appendix C.2 reports the full comparisons. The risk measure is persistent, with an adjacent-year correlation of 0.937 in the tax sample. Appendix C.3 examines period averages and longer differences. Persistence does not by itself establish that an imprecise estimate is caused by measurement error or limited power. I therefore interpret Fact 1 as a conditional association and distinguish the observed rate from a tax promised before investment.

2.3 Fact 2. Countries with Lower Risk Receive More FDI on Average.

The argument that investor protection expands the potential tax base has an empirical counterpart. Countries with lower average contract risk receive more net FDI relative to GDP. This is a relationship across countries, whereas Fact 1 uses changes within countries. Keeping that distinction explicit is necessary before interpreting the two patterns together.

Let f_{ht} denote net FDI inflows as a percentage of GDP and X_{ht} contain log real GDP per capita, log population, and natural resource rents. For country averages over jointly observed years, I estimate

$$\bar{f}_h = \alpha + \beta \bar{q}_h^L + \delta' \bar{X}_h + u_h, \quad (2)$$

where $\bar{q}_h^L$ averages the preceding calendar year's contract risk. Table 2 compares the same 136 countries before and after adding X . The coefficient changes from -3.995 to -3.056 percentage points of GDP per risk point. The controlled estimate is significant at the 10% level. Excluding the original haven group gives -1.424 in 116 countries, significant at the 5% level. Thus the controlled negative association does not arise solely from the inclusion of havens. The unadjusted association does weaken substantially when they are excluded, as the full comparison in Appendix C.4 shows.

This cross-country relationship describes persistent differences in destinations rather than the response to a change in risk within a country. Annual regressions do not establish the same

TABLE 2: Contract risk and net FDI inflows relative to GDP

	(1)	(2)	(3)
<i>Country averages</i>			
Contract risk	-3.995** (1.970)	-3.056* (1.597)	-1.424** (0.610)
Observations	136	136	116
Countries	136	136	116
Income, population and rents	No	Yes	Yes
Exclude original havens	No	No	Yes

Notes: The dependent variable is net FDI inflows as a percentage of GDP, including zero and negative flows. Contract risk is lagged one year. Columns (1) and (2) use identical observations. All columns use country averages and HC3 standard errors. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

investment response, and averaging over several years does not eliminate that distinction. Appendix C.3 compares the time variation explicitly. Appendix C.4 reports the annual regressions, the original log-flow outcome, and all haven exclusions. The model concerns investment chosen in anticipation of taxation, whereas net FDI also includes financial transactions among related firms.

2.4 Fact 3. Low-Tax Countries Have Larger Profit Than Employment Shares.

The first two facts concern investment and the tax rate, but the government taxes reported profits. Table 3 shows why these objects must be separated. In 2021, hosts with statutory rates below 15% account for 13.44% of profitable groups' reported profits and 3.01% of their employment on identical reporting cells. Including all profit statuses gives 16.76% and 2.97%. In both populations, the profit share exceeds the employment share in every year from 2017 to 2021.

TABLE 3: Profit and employment shares in hosts with statutory rates below 15%

Year	Profitable groups			All profit statuses		
	Profits	Employment	Difference	Profits	Employment	Difference
2017	12.52	2.13	10.39	15.07	2.32	12.76
2018	9.39	2.01	7.38	14.71	2.29	12.41
2019	10.21	2.34	7.88	11.87	2.49	9.38
2020	11.76	2.09	9.67	22.79	2.31	20.48
2021	13.44	3.01	10.43	16.76	2.97	13.79

Notes: Shares and differences are in percentage points. Profits and employment use identical cells. The balanced samples contain 1,008 pairs for profitable groups and 1,487 for all profit statuses. The latter retains negative aggregate profits. Reported zero employment is included.

The ordering survives the alternative tax thresholds, reporting samples, and individual country exclusions in Appendix C.6. The profit share is not itself an estimate of shifted profits. Productivity, industry composition, dividend inclusion, and reporting conventions can also separate profits from employment (OECD, 2023). The observation establishes that the location of productive activity does not mechanically determine the location of the reported tax base. That distinction motivates allowing firms to adjust reported profits after capital is installed.

The direct estimates of profit shifting provide a separate check on whether risk systematically predicts the location of profits. Their negative association with risk survives controlling for statutory taxes alone but weakens after adding income, population, and resource rents. Appendix C.4 retains the complete regressions. I therefore use the persistent geographic separation of profits and employment to motivate profit mobility, without treating the risk–shifting correlation as an independently established effect of reputation.

Taken together, the facts motivate a question about tax announcements. If better investor protection attracts capital, why might it accompany a lower tax? A larger investment base is only part of the government’s problem. The response of investment to the promise and the mobility of reported profits also affect what the government can collect. The next section gives these decisions explicit timing. Because observed statutory rates need not be promises made before investment, this theory explains a possible mechanism behind the motivating patterns rather than identifies their cause. Appendix C.7 sets out the observations needed for a direct test.

3 Model

The empirical analysis separates foreign investment from the profits reported in a host country. The model gives these margins different timing. Firms install productive capital before the government implements its tax but choose where to report profits afterward. This allows me to study whether a better reputation leads the government to tax an enlarged investment base more heavily or to reduce the announcement because investment responds more strongly to a credible promise.

3.1 Environment

Begin with a one-period economy containing a unit measure of foreign-owned atomistic firms, each with a capital endowment $\bar{k}$. Each firm allocates k to the host and $\bar{k} - k$ to a foreign location. Profits before shifting are $\rho(k)$ and $\rho(\bar{k} - k)$, where $\rho(0) = 0$ and ρ is increasing and strictly concave on the investment domain. The foreign tax rate is $\tau^* \in (0, 1)$. The host tax rate lies in $[0, 1]$. The foreign rate is fixed throughout each reputation comparison.

The government first announces a host tax rate τ^R . Firms observe the announcement and choose investment. The government then implements the tax τ , after which firms choose a reporting transfer θ , positive for inward shifting. This timing distinguishes sunk real investment from the profits that remain adjustable after the policy becomes known. All tax announcements and implemented rates are public. Section 4.2 extends this timing to two investment cohorts with no capital carried between them.

The government maximizes host tax revenue. Its privately known type is fixed. The two-period extension discounts revenue in the second period by $\beta \in (0, 1)$. A commitment type must implement its announcement, while an opportunistic type can revise the rate after investment. Both choose announcements to maximize revenue. Before observing any policy, investors assign

a probability of $p \in (0, 1)$ to the commitment type. An opportunistic government can revise its tax but may instead honor its announcement to preserve future revenue.

The objective isolates the fiscal consequences of limited commitment. Domestic factor incomes, public-good preferences, and domestic ownership are outside the model, so the revenue comparison is not a national welfare ranking.

3.2 Timing and Decisions

Solving backward, firms' reporting choices determine the tax base available after investment. The government chooses its implemented rate in response to that base. Firms anticipate this tax decision when allocating capital, and each government type anticipates their investment when announcing its policy.

Stage 4. Firms' Profit-Shifting Problem. After observing the implemented host tax τ , each firm chooses the reporting transfer θ , taking its installed capital k as given. In the baseline technology, shifting uses real resources in both locations and those costs are tax deductible. When $0 < k < \bar{k}$, the firm solves

$$v(k, \tau, \gamma) = \max_{\theta} (1 - \tau) \left[\rho(k) + \theta - \frac{\gamma \theta^2}{2\rho(k)} \right] + (1 - \tau^*) \left[\rho(\bar{k} - k) - \theta - \frac{\gamma \theta^2}{2\rho(\bar{k} - k)} \right], \quad (3)$$

subject to both taxable profits in brackets being nonnegative. The parameter $\gamma > 0$ measures the curvature of resource costs associated with organizing, documenting, and maintaining arrangements for shifting profits. A permanent increase in γ represents a restriction that makes shifting more costly. Fines paid to the government would require different revenue accounting. Alternative cost incidence and deductibility are examined in Appendix B.3.

The objective is strictly concave in the transfer. Before imposing the constraints requiring nonnegative tax bases, its maximizer is

$$\theta^u(\tau, k, \gamma) = \frac{(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)}{\gamma[(1 - \tau)\rho(\bar{k} - k) + (1 - \tau^*)\rho(k)]}. \quad (4)$$

If this transfer would make either taxable profit negative, the firm instead chooses the nearest transfer satisfying both nonnegativity constraints. Denote its choice by $\theta^*(\tau, k, \gamma)$. Appendix A.1 derives the feasible transfers. When all capital is in one location, shifting is zero, consistent with the limiting choice as investment approaches that location. Host revenue at aggregate investment K is

$$R(\tau, K, \gamma) = \tau \left[\rho(K) + \theta^*(\tau, K, \gamma) - \frac{\gamma \theta^{*2}(\tau, K, \gamma)^2}{2\rho(K)} \right]. \quad (5)$$

The tax calculation therefore incorporates the nonnegativity constraints on taxable profits. If the host tax base is exhausted, its revenue is zero.

Stage 3. The Opportunistic Government's Tax Problem. The commitment type implements its announced rate τ^R . The opportunistic type chooses the implemented rate after aggregate investment K is sunk, anticipating firms' reporting choices. Its tax response is

$$\tau^O(K, \gamma) \in \arg \max_{\tau \in [0,1]} R(\tau, K, \gamma). \quad (6)$$

The superscript R denotes the announced rate, which the commitment type must honor. The rate τ^O instead maximizes current revenue after investment. The symbol τ denotes the rate actually implemented, and τ^* denotes the foreign rate. An announcement affects τ^O through the investment it attracts. In the two-period model, the opportunistic type may choose τ^R instead of τ^O to preserve future revenue.

Stage 2. Firms' Investment Problem. Firms allocate capital after observing the announcement but before learning the implemented tax. At a common announcement by both types, beliefs remain at p . An individual firm takes each possible implemented rate as given and solves

$$\max_{k \in [0, \tilde{K}]} \left\{ p v(k, \tau^R, \gamma) + (1 - p) v(k, \tau^O(K, \gamma), \gamma) \right\}. \quad (7)$$

In a symmetric equilibrium the chosen k equals aggregate investment K . This consistency condition determines $K(\tau^R, p, \gamma)$. An individual firm does not internalize the response of the aggregate tax to its own investment. When firms anticipate τ^R with certainty, investment is denoted by $\tilde{K}$ and satisfies

$$\tilde{K}(\tau^R, \gamma) = \arg \max_{k \in [0, \tilde{K}]} v(k, \tau^R, \gamma). \quad (8)$$

The shifting cost structure and strict concavity of ρ make the firm's optimized payoff strictly concave in investment. The firm consequently has a unique, continuous investment response. Appendix A.2 establishes this property including the base constraints.

Stage 1. Governments' Announcement Problems. Each type chooses its announcement anticipating the beliefs, investment, and tax choices that follow. At a common announcement, the commitment type receives $R(\tau^R, K(\tau^R, p, \gamma), \gamma)$ and the opportunistic type receives $R(\tau^O(K, \gamma), K, \gamma)$. An unexpected announcement can change beliefs, so it cannot be evaluated by holding p fixed. Section 4.1 specifies those beliefs and checks that neither type gains by announcing a different rate. Among the common announcements that neither type wants to change, I study the one yielding the highest revenue to the commitment type.

The Full-Commitment Benchmark. When $p = 1$, the government's announcement maximizes $R(\tau^R, \tilde{K}(\tau^R, \gamma), \gamma)$. If the optimal rate lies strictly between zero and one, it satisfies

$$R_\tau(\tau^R, \tilde{K}, \gamma) + R_K(\tau^R, \tilde{K}, \gamma) \tilde{K}_{\tau^R} = 0. \quad (9)$$

The second term is the investment effect that the government ignores once capital is sunk. When the reporting choice is differentiable, the envelope theorem gives $R = -\tau v_\tau$ and hence $R_K = -\tau v_{k\tau}$. Thus, when $v_{k\tau} < 0$, a higher tax lowers investment and additional investment raises revenue. At the optimal announcement, the government would consequently gain revenue from increasing the tax if it could hold investment fixed. Lemma 1 shows that revenue

is strictly concave in the implemented tax wherever neither reporting constraint binds. When both tax choices are in that interval, it follows that $\tau^O > \tau^R$. The government would prefer a higher tax after investment than it would promise beforehand.

4 Analysis

The government chooses its tax promise anticipating how much investment that promise will attract. Better reputation changes this choice by changing investors' expectations about implementation. I first characterize the effect when an opportunistic government only values current revenue, and then allow it to preserve a reputation for future investors. The numerical results follow each model's analytical results so that the tax, investment, and reporting choices can be interpreted together.

4.1 The One-Period Model

The full-commitment benchmark explains why the government would prefer to choose taxes before investment. It does not determine how the optimal announcement changes as that commitment becomes more credible. I now vary the probability that the government must honor its promise. With no future revenue at stake, the opportunistic type chooses its current best response. This isolates the effect of reputation on the investment base and its sensitivity to the announcement.

4.1.1 Investment and Equilibrium

Consider an equilibrium in which both government types announce the same rate τ^R , commonly called a pooling equilibrium. Observing the announcement then gives firms no information about the government's type, so they continue to assign probability p to commitment. The commitment type implements τ^R , while the opportunistic type chooses $\tau^O(K, \gamma)$ after investment. The first-order condition for firms' investment problem (7), evaluated at symmetric investment, is

$$pv_k(K, \tau^R, \gamma) + (1 - p)v_k(K, \tau^O(K, \gamma), \gamma) = 0. \quad (10)$$

An individual firm takes both taxes as given. The government's response to aggregate investment enters when differentiating the equilibrium condition, not when the individual firm chooses its capital.

To characterize the investment response, consider a differentiable solution to (10) with $0 < K < \bar{k}$ and $0 < \tau^O < 1$. Where the reporting constraints remain nonbinding, implicit differentiation gives, suppressing γ ,

$$\begin{aligned} K_p &= -\frac{v_k(K, \tau^R) - v_k(K, \tau^O)}{pv_{kk}(K, \tau^R) + (1 - p)[v_{kk}(K, \tau^O) + v_{k\tau}(K, \tau^O)\tau_K^O]}, \\ K_{\tau^R} &= -\frac{pv_{k\tau}(K, \tau^R)}{pv_{kk}(K, \tau^R) + (1 - p)[v_{kk}(K, \tau^O) + v_{k\tau}(K, \tau^O)\tau_K^O]}. \end{aligned} \quad (11)$$

Here $\tau_K^O = \partial \tau^O / \partial K$. A subscript τ^R denotes differentiation with respect to the announcement. If the denominator is negative, $\tau^R < \tau^O$, and $v_{k\tau} < 0$ between the two rates, a better reputation increases investment at a fixed announcement while a higher announcement reduces it. Better reputation then mitigates the investment distortion from anticipated taxation. These signs do not establish how the revenue-maximizing announcement changes.

The investment response determines what each type earns from a common announcement. To sustain that announcement, both types must also prefer it to an unexpected rate that may change investors' beliefs. Consider first the investment and tax choices when investors know the government is opportunistic. Write these choices as K_0 and τ_0 , and the resulting revenue as R_0 . They satisfy

$$K_0 \in \arg \max_k v(k, \tau_0, \gamma), \quad \tau_0 \in \arg \max_{\tau} R(\tau, K_0, \gamma). \quad (12)$$

Appendix A.3 proves that at least one such equilibrium exists under the baseline shifting technology. Fix one if there are several. The subscript zero indicates that investors assign no probability to commitment. At any other investment level, $\tau^O(K, \gamma)$ still denotes the rate maximizing current revenue.

To construct the equilibrium with a common announcement, assume that investors regard a government making any unexpected announcement as opportunistic with certainty. They then invest K_0 . Neither government type can earn more than R_0 from that alternative announcement. A common announcement can therefore be sustained if each type earns at least R_0 by making it. Appendix A.4 verifies the required choices and beliefs. This assumption about unexpected announcements supports the equilibria studied here. Other beliefs may support different outcomes, including announcements that reveal the government's type.

Among the common announcements satisfying these two incentive constraints, I study the one maximizing revenue when the promise is honored. Write it as $\hat{\tau}^R(p, \gamma)$, with the hat distinguishing the optimal announcement from an arbitrary proposed rate. For the derivatives below, suppose the optimal announcement is a strict local maximum in $(0, 1)$, investment and tax choices are differentiable, and both government types strictly prefer the common announcement to an unexpected one.

4.1.2 Reputation and the Tax Base

The tax base when the announcement is honored is $R(\tau^R, K(\tau^R, p, \gamma), \gamma) / \tau^R$. Its elasticity with respect to the announcement, including investment and reporting responses, is

$$\eta(\tau^R, p, \gamma) = -\tau^R \frac{\partial}{\partial \tau^R} \log \left[\frac{R(\tau^R, K(\tau^R, p, \gamma), \gamma)}{\tau^R} \right]. \quad (13)$$

At a revenue maximum with a tax strictly between zero and one, $\eta = 1$. Differentiating that condition gives

$$\frac{d\hat{\tau}^R}{dp} = -\frac{\eta_p}{\eta_{\tau^R}}, \quad \eta_{\tau^R} > 0. \quad (14)$$

The inequality is the strict second-order condition. A larger base at a fixed tax is therefore insufficient to sign the optimal announcement. The tax falls when reputation makes the base

more sensitive to the announcement. If reputation merely multiplied the entire base schedule by a positive factor, it would leave the optimal tax unchanged. What matters is how reputation changes the shape of that schedule.

The change in the optimal announcement can be decomposed into these two effects at any p where the choices are differentiable. Let $H(\tau^R, p) = R(\tau^R, K(\tau^R, p, \gamma), \gamma)$ denote revenue when the promise is honored, holding the shifting cost parameter fixed. Suppose $H_{\tau^R \tau^R} < 0$ at the optimum, the rate lies in $(0, 1)$, and both types strictly prefer the common announcement. Then

$$\frac{d\hat{\tau}^R}{dp} = - \frac{\overbrace{(R_{\tau K} + R_{KK} K_{\tau^R}) K_p}^{\text{investment expansion}} + \overbrace{R_K K_{\tau^R p}}^{\text{change in investment response}}}{H_{\tau^R \tau^R}}. \quad (15)$$

Differentiating $H_{\tau^R} = R_{\tau} + R_K K_{\tau^R} = 0$ gives this expression. Because $H_{\tau^R \tau^R} < 0$, the numerator determines whether the optimal announcement rises or falls. The first term captures the effect of additional investment at a given announcement. The second captures the change in investment's response to that announcement. When the first term is nonnegative and $R_K > 0$, the announcement falls only if the stronger investment response outweighs the incentive to tax the larger base. Equation (15) identifies what must be checked in the numerical comparisons, rather than assigning the same sign to every economy. Appendix B.3 reports these components for the cost comparisons in Section 5.1. If a tax reaches zero or one, or the optimal choice changes discontinuously, revenue must be compared directly instead of using this derivative.

The comparison becomes particularly clear when investors initially believe the government is opportunistic with certainty. Evaluate derivatives at the equilibrium (K_0, τ_0) defined above, with $0 < K_0 < \bar{k}$, $0 < \tau_0 < 1$, and neither reporting constraint binding for nearby choices. Lemma 1 establishes $R_{\tau\tau} < 0$ and $R_K = -\tau v_{k\tau}$ under the baseline technology. Suppose v and R are three times continuously differentiable near this equilibrium.

Proposition 1 (Reputation, investment, and the announced tax). *If $v_{k\tau} \neq 0$ and $v_{kk} + v_{k\tau} \tau_K^O < 0$, there exist pooling weak perfect Bayesian equilibria for sufficiently small $p > 0$ in which the choices vary differentiably with p from (K_0, τ_0) . The announcement is chosen to be a strict local maximum of revenue when the promise is honored. As p increases within this set of equilibria, the announcement falls, while revenue and the tax base when the promise is honored rise. In particular,*

$$\left. \frac{d\hat{\tau}^R}{dp} \right|_{p=0} = - \frac{\tau_0 (v_{k\tau})^2}{R_{\tau\tau} (v_{kk} + v_{k\tau} \tau_K^O)} < 0, \quad (16)$$

$$\left. \frac{dK(\hat{\tau}^R(p), p)}{dp} \right|_{p=0} = 0, \quad \left. \frac{d^2 K(\hat{\tau}^R(p), p)}{dp^2} \right|_{p=0} = - \frac{2v_{k\tau}}{v_{kk} + v_{k\tau} \tau_K^O} \left. \frac{d\hat{\tau}^R}{dp} \right|_{p=0}. \quad (17)$$

Investment therefore increases if $v_{k\tau} < 0$ and decreases if $v_{k\tau} > 0$ for sufficiently small $p > 0$.

When $p = 0$, investors know the government is opportunistic and ignore its announcement. The tax maximizing revenue if the promise were honored is then τ_0 , the same rate the opportunistic type would choose after investment. A small increase in p gives the announcement some influence on investment. At the optimal announcement, the change in this investment response is proportional to the initial increase in p , while the change in investment itself is of smaller order because the two possible tax rates initially coincide. The government therefore has an incentive to lower its announcement before the expansion of its investment base creates

a comparable incentive to raise it. If a higher tax lowers the marginal return to investment, the lower announcement also attracts more capital. This explains how the tax and investment patterns motivating Facts 1 and 2 can arise together. The argument does not require a negative relationship between risk and profit shifting in the data.

The condition $v_{kk} + v_{k\tau}\tau_K^O < 0$ means that additional aggregate investment lowers a firm's marginal return after accounting for the government's tax response. The tax result requires investment to respond to taxation, but allows either sign for that response. The stronger conclusion that investment rises requires $v_{k\tau} < 0$. The assumptions on tax rates and reporting constraints matter because confiscation or a binding limit on reported profits can change the government's incentives. Appendix B.1 extends the tax result to firms with different investment responses. A mean reporting elasticity by itself cannot establish these conditions.

These conditions are not confined to one calibrated economy. For $\rho(k) = k^\alpha$, $\bar{k} = 2$, and $\gamma = \tau^*/[2(1 - \tau^*)]$, the equilibrium at $p = 0$ has $K = 1$ and $\tau = \tau^*$. For any $0 < \alpha < 1$ and $0 < \tau^* < 1$, equation (16) gives

$$\left. \frac{d\hat{\tau}^R}{dp} \right|_{p=0} = -\frac{\alpha(\tau^*)^2}{(1 - \alpha)(4 - \tau^*) + \alpha\tau^*} < 0. \quad (18)$$

All the required inequalities are strict, so the result survives small changes in the model parameters. A small increase in γ raises τ_0 above τ^* . Firms then shift profits outward under either government type for sufficiently small $p > 0$. Appendix A.5 gives the derivation. The falling announcement therefore does not depend on one parameter choice or on profits being shifted into the host country.

4.1.3 An Economy without Profit Shifting

Without shifting, the opportunistic type chooses a tax of one whenever the host has positive profits. Investors consequently choose capital according to

$$p(1 - \tau^R)\rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (19)$$

For a general profit function, define

$$g(K) = \frac{-\rho(K)\rho''(K)}{\rho'(K)^2} + \frac{\rho(K)}{\rho'(K)} \frac{-\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)}. \quad (20)$$

This index combines the curvature of returns in the host and foreign locations. It determines the announcement at which the marginal revenue from a higher rate equals the revenue lost through investment. For the following result, let ρ be three times continuously differentiable on $(0, \bar{k}]$, with $\rho' > 0$, $\rho'' < 0$, and $\rho'(0+) = \infty$.

Proposition 2 (The immobile-profit benchmark). *If g is strictly increasing and bounded near zero, the one-period revenue-maximizing announcement without shifting is unique for every $0 < p \leq 1$ and satisfies*

$$\hat{\tau}^R(p) = \frac{g(K(p))}{1 + g(K(p))}. \quad (21)$$

Both investment and the announced tax increase strictly with p . This allocation is supported by a pooling weak perfect Bayesian equilibrium when investors regard any government making an unexpected announcement as opportunistic with certainty. For $\rho(k) = zk^\alpha$, $z > 0$ and $0 < \alpha < 1$, the conditions hold and the announcement reduces to

$$\hat{\tau}^R(p) = \frac{1 - \alpha}{1 - \alpha K(p)/\bar{k}}. \quad (22)$$

Appendix A.6 proves the result by combining the firm's investment condition with the government's tax condition. When investors know the government is opportunistic, anticipated confiscation eliminates host investment. The government therefore starts without the positive investment base underlying Proposition 1. Allowing profits to move after investment changes this starting point and can change the tax response to reputation. It does not imply that any positive amount of shifting reverses the response at every level of reputation.

The numerical production function $\rho(k) = zk^\alpha - r^*k$, with $r^* \geq 0$, belongs to the same class when $\alpha z \bar{k}^{\alpha-1} > (2 - \alpha)r^*$. Appendix A.6 derives the increasing g from this restriction on production parameters. The benchmark parameters satisfy it, so the static comparison does not rely on dropping the rental deduction only in the economy without shifting. Concavity of profits alone need not make g increase, so the result continues to require the stated restriction on the curvature of investment returns.

For interpreting observed rates, the announcement must also be distinguished from the expected implemented tax. Without shifting, the latter is $1 - p(1 - \tau^R(p))$. The firm's condition gives $p(1 - \tau^R(p)) = (1 - \tau^*)\rho'(\bar{k} - K(p))/\rho'(K(p))$, which increases with investment. Thus the expected implemented tax falls with reputation in the benchmark without shifting even though its announcement rises. A negative relationship between reputation and taxes across countries cannot distinguish the models unless the observed rate is known to be the announcement made before investment.

The numerical solution below compares these two announcement schedules using the same production parameters. It also separates investment from the reported tax base, making clear how profit mobility changes the government's incentive to tax an investment that has already been installed.

4.1.4 Numerical Results

Proposition 1 shows that a small improvement from very low reputation lowers the optimal announcement and, when $v_{k\tau} < 0$, raises investment. The numerical exercise asks whether these responses persist over larger changes in reputation. I vary p while holding the production and shifting technologies fixed. For each value, I solve firms' investment choices together with the opportunistic government's tax choice, then find the common announcement yielding the

highest revenue to the commitment type while respecting both types' announcement incentives. Figure 1 compares the announced rate τ^R , investment K , and the rate τ^O chosen after investment with the corresponding outcomes under the same production parameters without shifting.

Parameter Values. The benchmark uses $\rho(k) = k^{0.66} - 0.04k$, $\bar{k} = 10$, $\tau^* = 0.15$, and $\gamma = 1.25$. These values define an illustrative economy. They do not estimate a representative country, and no risk score is converted into the probability of commitment p . Both reporting economies use the same production parameters. Section 5.1 relates the reporting cost parameter to estimates in the literature and varies the technology. Appendix B.4 describes the solution and incentive checks.

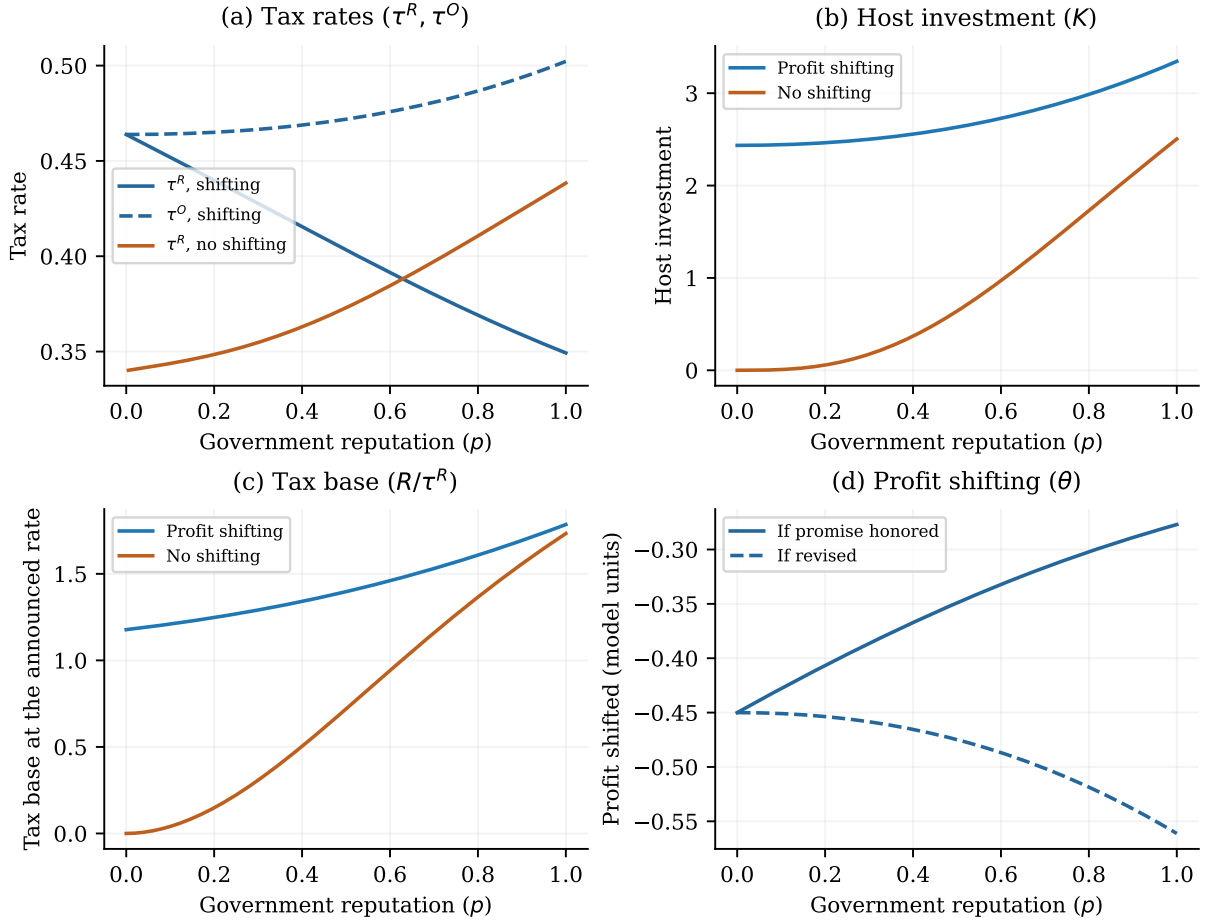


FIGURE 1: Taxes, investment, and reported profits in the one-period model

Notes: All panels report the one-period model with benchmark production parameters and $\gamma = 1.25$. Panel (c) evaluates R/τ^R when the announcement is honored. Panel (d) shows the transfer θ in model units. Negative values denote outward shifting. The no-shifting tax plotted is the announcement. Its implemented tax under the opportunistic type is one.

Tax Rates, Investment, and the Tax Base. In the shifting economy, Figure 1(a)–(c) shows that increasing p from 0.1 to 0.9 lowers τ^R from 0.452 to 0.359, while K rises from 2.442 to 3.153. When the announcement is honored, the tax base rises from 1.210 to 1.693 and revenue from 0.547 to 0.607. The larger base therefore accompanies a lower revenue-maximizing announcement.

Better reputation makes the announcement more consequential for investment, giving the government a stronger incentive to promise a lower rate.

Panel (d) shows why the distinction between the announced and implemented taxes matters. When the announcement is honored, the magnitude of outward shifting falls from 0.428 to 0.289 over the same increase in reputation. When the opportunistic government revises it, outward shifting instead rises from 0.451 to 0.538, as τ^O increases from 0.464 to 0.494. The larger investment base makes a higher tax attractive after capital is installed, even though the government promises a lower rate beforehand to attract that capital. Appendix B.5 reports the tax rates and the associated investment, tax base, and revenue.

These results show why the location of investment and the location of taxable profits must enter the analysis separately. A more credible promise attracts capital, but firms can still move reported profits when the tax is implemented. The government consequently announces a lower rate even while its tax base expands. The geographic separation of profits and employment in Section 2.4 motivates this distinction. It does not measure the response to a tax revision shown in the figure.

The one-period comparison takes the opportunistic government's revision as given. The next part asks whether the same government would revise its tax when doing so sacrifices future investment. Cost assumptions and limits on profit mobility are examined together in Section 5, after establishing this additional incentive.

4.2 The Two-Period Model

The one-period analysis isolates how beliefs change after an announcement when the opportunistic type reoptimizes following investment. Over two periods, that type can honor its first promise to preserve future revenue. The probability of a commitment type and the probability that the announcement is honored then differ. The model must therefore allow the government to honor or revise its announcement.

4.2.1 Equilibrium

In each period, a new cohort makes the investment and reporting decisions in Section 3. The government's type is fixed, and investors observe its earlier announcements and implemented taxes. Let $\sigma \in [0, 1]$ be the probability that the opportunistic type honors its first announcement. Investors then expect the promise to be honored with probability π . After observing that it was honored, they assign probability q to the commitment type. Bayes' rule gives

$$\pi = p + (1 - p)\sigma, \quad q = \frac{p}{\pi}. \quad (23)$$

Only the opportunistic type can implement a tax different from its announcement. A revision, therefore, reveals its type. Honoring the announcement instead raises the probability that investors assign to commitment when $\sigma < 1$. When both types always honor it, the observation conveys no information, and that probability remains p . In evaluating whether the government would change its decision, the beliefs associated with each observed action are held fixed.

Second-Period Decisions. For each belief q , take an equilibrium from Section 4.1 in which both types make the same announcement. Write $R_2^R(q, \gamma)$ and $R_2^O(q, \gamma)$ for the resulting second-period revenues of the commitment and opportunistic types. The subscript denotes the period, and the superscript denotes the type. If investors know the government is opportunistic, $q = 0$, and both revenues equal $R_0(\gamma)$. Thus, the future revenue at stake is determined by the same investment and taxation problem as the current revenue.

First-Period Investment and Reporting. First-period investment is $K(\tau^R, \pi, \gamma)$, obtained from (10) with π in place of p . Use a tilde for current revenue when the announcement is implemented and a hat for its maximum after revision. These revenues are

$$\begin{aligned}\tilde{R}(\tau^R, \pi, \gamma) &= R(\tau^R, K(\tau^R, \pi, \gamma), \gamma), \\ \hat{R}(\tau^R, \pi, \gamma) &= \max_{\tau} R(\tau, K(\tau^R, \pi, \gamma), \gamma).\end{aligned}\tag{24}$$

Both revenues hold the same sunk investment fixed. The notation $\tilde{R}$ does not presume that firms anticipated that the tax promise would be honored with probability one. Only when $\pi = 1$ does this investment equal $\tilde{K}$ in (8).

The Opportunistic Government's Tax Decision. After first-period investment, the opportunistic type compares current revenue when the announcement is honored with the maximum from revision, including the second-period revenues following each action. Define its net gain from honoring the announcement by

$$\mathcal{G}(\tau^R, \sigma; p) = \tilde{R} - \hat{R} + \beta [R_2^O(q, \gamma) - R_0(\gamma)].\tag{25}$$

Raising the tax increases current revenue by $\hat{R} - \tilde{R}$, but gives up the discounted revenue difference $\beta[R_2^O(q, \gamma) - R_0(\gamma)]$. The government honors its announcement when the latter is at least as large. A low promised rate is attractive before investment because it draws in capital, yet can be costly to maintain afterward because it leaves more current revenue to be gained from revision. This incentive constraint connects the tax and investment comparisons. A government attracts capital through the tax investors expect it to implement, so an attractive announcement alone cannot account for the investment associated with better protection in Fact 2.

First-Period Announcement Incentives. Both types choose their announcements before investment. As in the one-period model, assume that investors regard an unexpected announcement as coming from the opportunistic type with certainty. Each type must prefer the proposed common announcement to such an alternative. The following conditions combine that requirement with the incentive to honor or revise the tax after investment.

Proposition 3 (Government tax choices and beliefs). *Fix second-period equilibria as described above and $0 < p < 1$. Consider a common first-period announcement and a probability σ with which the opportunistic type honors it. A revision must implement a rate different from the announcement. Assume that investors regard any government making an unexpected announcement as opportunistic with certainty. The proposed choices constitute a pooling weak perfect Bayesian equilibrium if and only if firms optimize, beliefs satisfy (23), and*

$$\begin{cases} \mathcal{G} \leq 0, & \sigma = 0, \\ \mathcal{G} = 0, & 0 < \sigma < 1, \\ \mathcal{G} \geq 0, & \sigma = 1, \end{cases} \quad (26)$$

subject to the two announcement constraints

$$\begin{aligned} \tilde{R} + \beta R_2^R(q, \gamma) &\geq (1 + \beta)R_0(\gamma), \\ \sigma[\tilde{R} + \beta R_2^O(q, \gamma)] + (1 - \sigma)[\hat{R} + \beta R_0(\gamma)] &\geq (1 + \beta)R_0(\gamma). \end{aligned} \quad (27)$$

Appendix A.7 verifies choices at the announcement, tax implementation, and second-period stages. The proposition allows the opportunistic government to honor its announcement, revise it, or be indifferent and randomize between the two actions. Taxes of zero and one are also allowed. No additional derivative restrictions are needed beyond the existence of the optimal choices and second-period equilibria specified above. Among the equilibria satisfying these conditions, I study the one maximizing the commitment type's total discounted revenue. This comparison is within equilibria where both types make the same announcement and does not rule out other equilibria. The expected first-period tax revision is $(1 - \pi)[\tau^O(K, \gamma) - \tau^R]$, the expected difference between the implemented and announced rates on the same investment.

The simplest case is one in which both types always honor the announcement, so $\pi = 1$ and $q = p$. Consider the lowest announcement that the opportunistic type is willing to honor. Suppose it is indifferent between honoring and revising that rate, both types strictly prefer making this announcement to making an unexpected one, and a higher announcement reduces the current gain from revision. Differentiating the condition equating that gain to the loss of future revenue gives

$$\frac{d\tau^R}{dp} = \frac{\beta \partial R_2^O(p, \gamma) / \partial p}{\partial[\hat{R}(\tau^R, 1, \gamma) - \tilde{R}(\tau^R, 1, \gamma)] / \partial \tau^R}. \quad (28)$$

A larger loss of future revenue then allows the government to sustain a lower promised rate. This derivative concerns the lowest rate that is always honored. The equilibrium maximizing the commitment type's revenue may instead involve occasional revision, and its choices need not change continuously with reputation. When the opportunistic type randomizes, its current gain from revision equals its discounted loss of future revenue. A change in reputation can then affect both the frequency of honoring the promise and what investors infer from that action. The effect on the announcement must account for both adjustments.

A promise can be credible yet attract little investment. If the announcement already equals the rate maximizing revenue after investment, the government has no current gain from raising it. Without shifting, however, such a tax can leave investors with little reason to install capital. Appendix A.8 proves that, as p approaches zero, first-period revenue also approaches zero

whenever both types always honor their announcement. With shifting, firms can protect part of their profits from a tax increase, and positive investment and revenue can remain even when investors know the government is opportunistic. Whether the government sustains that investment by always honoring its promise or prefers a lower announcement with occasional revision is an equilibrium choice. Section 4.2.2 compares the resulting promised rates, the probabilities of honoring them, and investment. Appendix B.5 reports the numerical values, and Section 5.4 varies the importance of future revenue.

4.2.2 Numerical Results

The numerical solution computes the investment sustained by a tax promise under the incentive constraints in Proposition 3. For each initial probability of commitment, I jointly solve the announcement, investment, the opportunistic type's probability of honoring its promise, and the resulting beliefs. The search permits $\sigma = 0$, $0 < \sigma < 1$, and $\sigma = 1$. Figure 2 reports the first period at $\beta = 0.6$, with the same production and reporting parameters as Figure 1. The government now also values the revenue it can preserve in the second period.

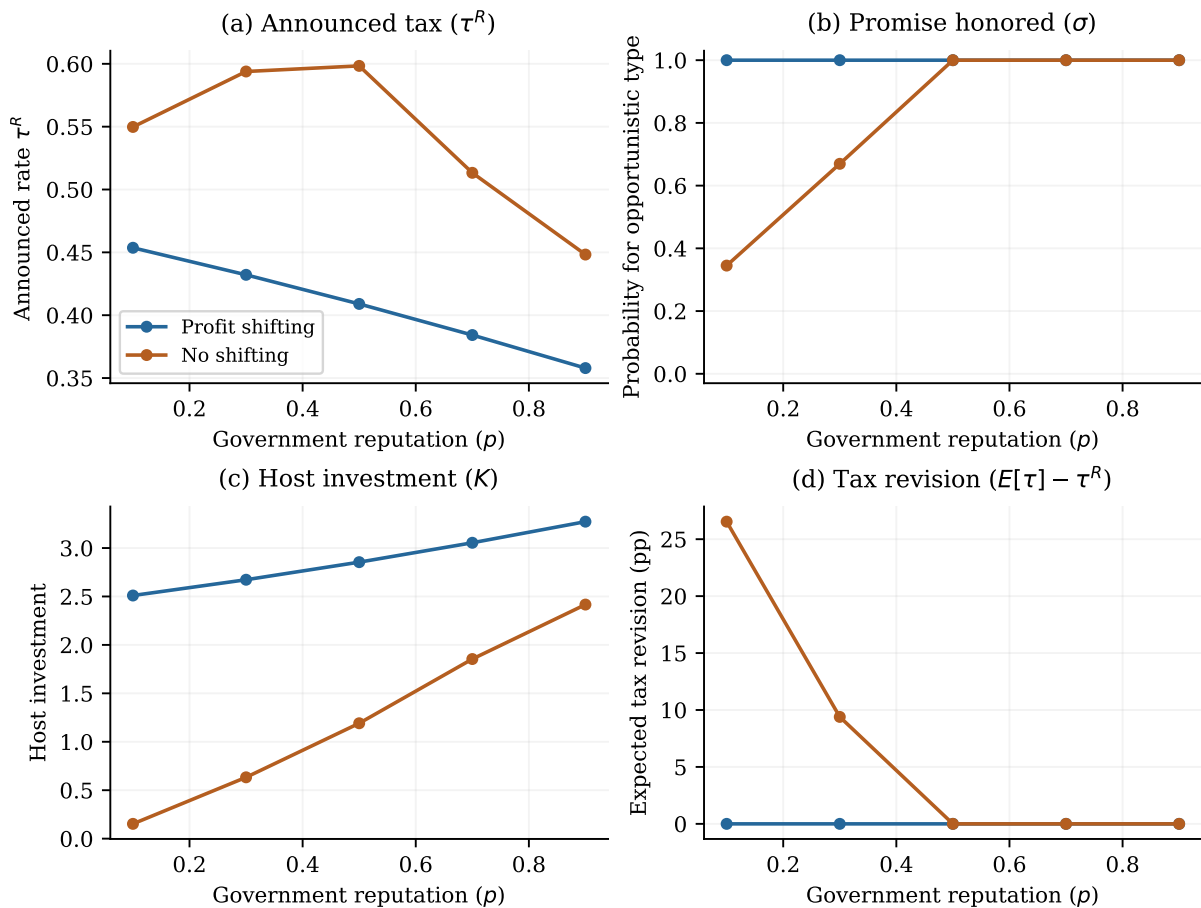


FIGURE 2: Tax policies and investment in the two-period model

Notes: Benchmark parameters and $\beta = 0.6$. All panels report the first period of the two-period model. In panel (b), σ is the opportunistic type's probability of honoring its announcement, not the initial probability p of the commitment type. The expected tax revision is $100(1 - \pi)[\tau^O(K, \gamma) - \tau^R]$, in percentage points. Markers show computed equilibria at the five displayed values of p . Lines connect those points.

With profit shifting, the opportunistic government honors the first announcement at all five reported levels of initial reputation, so $\sigma = 1$. As p rises from 0.1 to 0.9, the announced rate falls from 0.454 to 0.358, and investment rises from 2.510 to 3.272. The outcome at low initial reputation illustrates the difference from the one-period model. Investors assign a low probability to the commitment type, but they still expect the announced tax to be implemented because the opportunistic type prefers to preserve future revenue. Profit mobility also limits the revenue it could obtain by raising the current tax.

Without shifting, that investment can be taxed away once installed. At $p = 0.1$, the opportunistic government instead randomizes, with $\sigma = 0.345$. Investors then assign probability $\pi = p + (1 - p)\sigma = 0.411$ to the announced tax being honored. Investment is only 0.153, and the expected upward tax revision is 26.54 percentage points. Restricting the equilibrium search to announcements the opportunistic type is willing to honor with probability one instead selects a rate of 0.859 and an investment of only 0.071. A high announced tax reduces the temptation to raise it further but discourages the investment it is meant to attract.

At higher initial reputation, both types honor the announcement even without shifting, and the optimal announcement decreases. A falling tax schedule in the two-period model therefore does not distinguish the reporting technologies by itself. Their more consequential difference is how much investment can be sustained at low reputation and whether investors face a tax revision. This is the investment implication of the incentive constraint in Section 4.2. The comparison connects the model to the investment question raised by Fact 2. Two countries could announce similarly low rates yet attract different amounts of capital if investors assign different probabilities to subsequent revision. Statutory rates and broad contract risk alone do not identify that probability. Table 9 in Appendix B.5 reports the exact allocations, and Appendix B.4 explains the solution checks. Section 5.4 varies the value of future revenue to examine this incentive further.

5 Extensions and Sensitivity Checks

The analysis shows how a better reputation can lower the announced tax while attracting investment and how future revenue can induce the government to honor that announcement. Its scope depends on the investment response and the profits that remain mobile after the tax is chosen. The empirical facts distinguish investment from reported profits but do not identify these responses. I therefore vary the cost and timing of profit shifting, allow firms to differ, and examine the value of future revenue. The cost and mobility comparisons concern the one-period mechanism. The final exercise returns to the government's tax choice over two periods.

5.1 Profit-Shifting Costs

The cost parameter γ governs how strongly the reported base responds to the implemented tax. Holding investment fixed, I measure this response by the semi-elasticity

$$\varepsilon(\tau, K, \gamma) = -\frac{\partial \log[R(\tau, K, \gamma)/\tau]}{\partial \tau}. \quad (29)$$

At equal tax rates and equal profits before shifting, this is $1/[2\gamma(1 - \tau^*)]$. The benchmark gives 0.471. $\gamma \simeq 0.467$ gives 1.26 at the same reference point. With taxes expressed as fractions, this is approximately the percentage change in the base following a tax increase of one percentage point. It differs from η , which includes investment and concerns an announcement when investors are uncertain whether the government will honor it.

Using Compustat geographic segments, [Kallen \(2023\)](#) reports a mean firm-specific semi-elasticity of 1.26 and a mean inverse cost parameter of 1.00. The inverse cost parameter has a different definition from γ , so I compare the implied responses of reported profits rather than substituting that estimate directly. [Bilicka et al. \(2024\)](#) reports short- and long-run aggregate responses of 1.5 and 4.7 that also reflect positive-profit reporting and avoidance capability. [McClure \(2023\)](#) instead estimates the ratio of shifting costs to income. These estimates guide the numerical analysis but concern different choices made by firms. The comparison below isolates the response of reported profits while holding investment fixed, which the geographic shares in Section 2.4 do not identify.

The decline in Figure 1 extends the local result in Section 4.1 to a wider range of initial reputation in the benchmark economy. Whether it does so under other reporting technologies is a separate question. Equation (15) identifies the relevant comparison. Reputation expands investment at a given announcement, but it also changes its response to the announcement. The tax falls when the latter effect dominates. I vary reporting costs and the scope for adjustment to establish which parts of this comparison survive changes in the benchmark.

Table 4 holds the response of the tax base to a small tax change fixed at a common reference allocation and changes where shifting costs arise and whether they are deductible. This prevents a change in cost treatment from being confounded with an arbitrary change in the reference responsiveness of reported profits. The baseline incurs deductible costs in both countries. The alternatives remove deductibility or incur deductible costs only in the country from which profits are shifted. All rows solve the same one-period announcement problem as Figure 1.

I use reference semi-elasticities of 1.26 and 1.50, drawing respectively on [Kallen \(2023\)](#) and the recent meta-analytic estimate in [Beer et al. \(2018\)](#). The literature estimates need not hold investment fixed, so they guide the scale of the comparison rather than identify its cost parameters. Appendix B.3 derives the reporting rules and parameter choices for all three cost treatments.

The announced rate falls across the three levels of reputation in all six comparisons. At semi-elasticity 1.26, it falls from 31.5% to 27.2% with deductible costs in both countries, from 38.1% to 30.2% without deductibility, and from 30.6% to 27.9% with costs only at the origin. Cost treatment affects tax levels, but the decline does not depend on costs being deductible in both locations. At the middle allocation of the first row, the term for investment expansion in equation (15) is 0.076 and the term measuring the change in investment responsiveness is -0.841 . Their sum is negative. This calculation connects the tax decline to the mechanism in Section 4.1, rather than only to the ordering of three numerical tax rates. Appendix B.3 explains the decomposition and the associated cost parameters.

TABLE 4: Announced taxes in the one-period model under alternative shifting costs

Costs incurred in	Tax deductible?	Tax-base semi-elasticity	Announced tax rate (%)		
			$p = 0.1$	$p = 0.5$	$p = 0.9$
Both countries	Yes	1.26	31.5	29.3	27.2
Both countries	No	1.26	38.1	33.8	30.2
Country from which profits are shifted	Yes	1.26	30.6	29.3	27.9
Both countries	Yes	1.50	29.2	27.3	25.6
Both countries	No	1.50	34.3	30.9	28.0
Country from which profits are shifted	Yes	1.50	28.5	27.3	26.2

Notes: Entries are one-period equilibrium announcements τ^R in percent. Reputation p is the probability of the commitment type. Within each semi-elasticity group, production parameters and the reference response of the reported base are fixed. The reference holds investment fixed at equal tax rates and equal profits before shifting. Transfers satisfy nonnegative tax bases and face no additional limit.

5.2 Limits on Profit Mobility

A common response near equal taxes does not imply that firms can move equally large amounts of profit after a substantial tax increase. I therefore retain the first cost specification at semi-elasticity 1.26 and restrict the maximum transfer to a fraction of profits earned in the country of origin. Every positive limit leaves the reference response to a sufficiently small tax difference unchanged. A binding limit nevertheless makes the base less mobile when the government considers a large revision.

This restriction can overturn the decline. With a 5% transfer limit, the announced rate rises from 40.9% at $p = 0.5$ to 44.5% at $p = 0.9$, compared with a fall from 29.3% to 27.2% without an additional limit. Table 5 in Appendix B.3 reports all three limits, including cases in which the government chooses a tax of one. These comparisons explain why the reporting elasticity alone does not establish the mechanism. The analytical result requires the tax chosen by a government known to be opportunistic to lie strictly between zero and one, with differentiable investment responses. The numerical comparison also allows the government to confiscate all taxable profits. Similarly, the very high cost case $\gamma = 1000$ in Table 6 makes the announcement rise between intermediate and high levels of reputation. Table 8 in Appendix B.5 also varies production curvature jointly with shifting costs.

The transfer limit concerns arrangements already available when the tax changes. Creating a network or paying a fixed participation cost beforehand is a separate decision, as emphasized by Bilicka et al. (2024) and Altshuler et al. (2025). If only some firms incur those costs, the share of profits that remains mobile also matters. The next extension allows firms to differ in their investment and reporting opportunities.

5.3 Heterogeneous Firms and Tax Instruments

The evidence in Section 2.4 concerns aggregate profits and employment, while firms can differ in both investment responses and opportunities to shift profits. Appendix B.1 extends Proposition 1 to heterogeneous firms. The local decline in the announcement survives when the aggregate investment condition holds and at least one group responds to taxation. Each

group's investment rises or falls according to its own response. The tax conclusion therefore requires less than an increase in every firm's investment.

The analytical result still requires the tax chosen by a government known to be opportunistic to lie strictly between zero and one. A small group with mobile profits may not prevent confiscation when most profits cannot move. Appendix B.2 derives the required mobile share and shows why a common tax differs from separate taxes on mobile and immobile firms. Separate taxes allow the government to tax the immobile base more heavily without losing the mobile base. This increases the current gain from revising a common promise at fixed investment, although the full dynamic comparison also requires solving for future revenue again. Neither a mean reporting elasticity nor the geographic concentration of reported profits establishes that the government prefers such a rate to confiscation.

The incentive constraint also accommodates an external penalty for revision. Appendix B.2 shows that such a penalty raises the gain from honoring the announcement at fixed investment and second-period revenues. It separates this direct incentive effect from changes in the equilibrium announcement and investment.

5.4 The Value of Future Revenue

The two-period comparison depends on the future revenue the government would lose by revealing that it can revise its tax promise. Figure 3 varies β over 0.3, 0.6, and 0.9, holding the production and reporting technologies fixed. A government that values future revenue more is willing to forgo a larger current gain from revision. When the announced tax is the lowest rate it is willing to honor with certainty, this can allow a lower promise and attract more investment. When the government sometimes revises its announcement, both the frequency of revision and what investors infer from honoring the promise can change. The figure reports these decisions together.

At $p = 0.1$ without shifting, raising β from 0.3 to 0.9 increases the probability that the opportunistic type honors its promise from 0.270 to 0.397. Investment rises from 0.088 to 0.209. With shifting, the promise is honored at both discount factors and investment rises from 2.488 to 2.527. A greater concern for future revenue therefore reduces the loss of investment from anticipated revision in this comparison, but the investment difference between the reporting technologies remains large. All displayed allocations satisfy the announcement and implementation constraints in Proposition 3. Appendix B.4 gives the checks and their numerical tolerances.

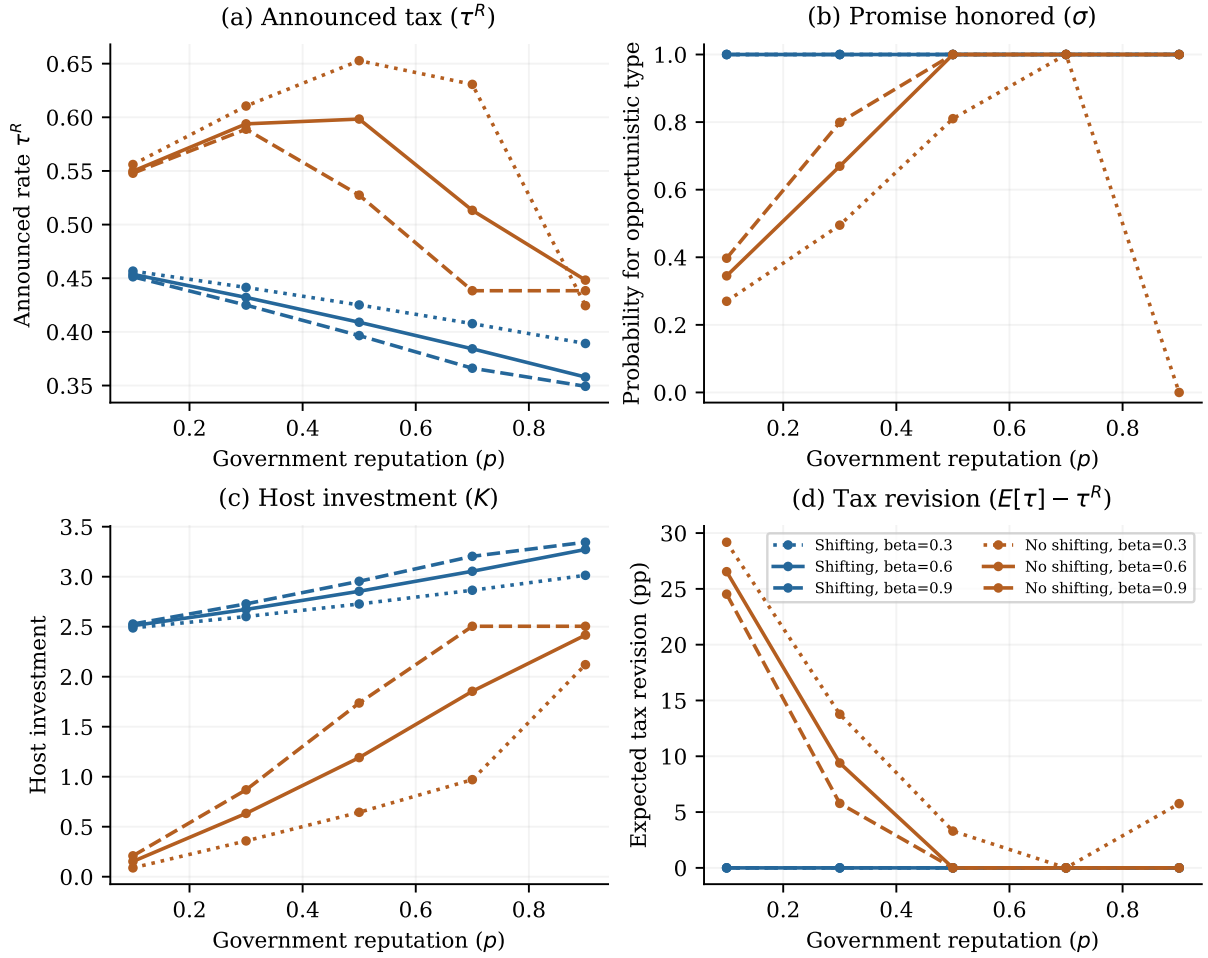


FIGURE 3: Tax policies in the two-period model at different discount factors

Notes: Blue lines allow profit shifting and orange lines exclude it. Solid, dotted, and dashed lines distinguish the three stated discount factors. Each marker reports a computed equilibrium at the displayed initial reputation. The expected tax revision refers to implemented and announced rates on identical investment, not to accounting tax rates.

6 Concluding Remarks

Better reputation can lead a government to promise a lower tax while attracting more investment. Credibility changes the response to a tax announcement as well as the size of the investment base. When investors expect the government to ignore its promise, announcing a lower rate attracts little additional capital. As the promise becomes credible, the same reduction becomes more effective. I prove that this incentive lowers the optimal announcement at low initial reputation under the stated conditions on taxation and investment. The result holds beyond a single choice of parameters. Investment expands when a higher tax lowers its marginal return. For larger improvements in reputation, the comparison also depends on the amount of profit firms can move after the tax is chosen.

Over two periods, these investment gains give the government a reason to honor its announcement. Revising the current tax sacrifices the future revenue associated with a reputation for commitment. Profit mobility can restrain the gain from revision and support investment at low initial reputation. A high tax may also be credible without profit shifting, but attract little capital. The relevant policy comparison is therefore the tax promise a government can sustain and the investment it attracts, rather than credibility considered separately from the promised rate.

The empirical evidence motivates this distinction without identifying the effect of a broken promise. Lower contract risk accompanies lower statutory rates within countries and greater FDI across countries, but these relationships use different variation. The persistence of the risk measure does not make them interchangeable. The geographic separation of profits and employment further shows why investment and the reported tax base should be distinguished. Testing the mechanism directly requires observing tax announcements before investment and their subsequent implementation. The model specifies how the investment response to those promises, the mobility of reported profits, and future revenue jointly determine the tax a government chooses.

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A Proofs

A.1 Constrained Profit Shifting

Solving the two quadratic nonnegative-base inequalities gives the feasible interval

$$\begin{aligned}\theta &\geq \max \left\{ \frac{\rho(K)(1 - \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - K)(-1 - \sqrt{1 + 2\gamma})}{\gamma} \right\}, \\ \theta &\leq \min \left\{ \frac{\rho(K)(1 + \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - K)(-1 + \sqrt{1 + 2\gamma})}{\gamma} \right\}.\end{aligned}\quad (30)$$

The interval contains zero. Since the firm's payoff is strictly concave in the transfer, its constrained optimum is the point in this interval closest to the unconstrained choice in (4). If profit before shifting is zero in either country, the transfer must be zero. This restriction follows from the shifting cost, which diverges for any nonzero transfer as the corresponding profit approaches zero.

Lemma 1 (Revenue When Reporting Constraints Do Not Bind). *When taxable profits are positive in both countries,*

$$R(\tau, K, \gamma) = \tau \rho(K) \left\{ 1 + \frac{1}{2\gamma} - \frac{1}{2\gamma} \left[\frac{(1 - \tau^*)[\rho(K) + \rho(\bar{k} - K)]}{(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)} \right]^2 \right\}. \quad (31)$$

For $0 < K < \bar{k}$, revenue is strictly concave in the tax over the interval where neither reporting constraint binds. Wherever the optimized reporting choice is differentiable,

$$R(\tau, K, \gamma) = -\tau v_\tau(K, \tau, \gamma), \quad R_K = -\tau v_{K\tau}. \quad (32)$$

For $\tau^* < \tau < 1$, both R_γ and $R_{\tau\gamma}$ are positive.

Proof. The reporting constraints do not depend on the tax rate. The envelope theorem therefore gives v_τ equal to minus the optimized host tax base. Multiplying by $-\tau$ and differentiating with respect to investment proves (32). Substitution of (4) into the host revenue function gives (31). Direct differentiation yields

$$\begin{aligned}R_\gamma &= \frac{\tau \rho(K)}{2\gamma^2} \left\{ \left[\frac{(1 - \tau^*)[\rho(K) + \rho(\bar{k} - K)]}{(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)} \right]^2 - 1 \right\}, \\ R_{\tau\gamma} &= \frac{\rho(K)}{2\gamma^2} \left\{ \left[\frac{(1 - \tau^*)[\rho(K) + \rho(\bar{k} - K)]}{(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)} \right]^2 \right. \\ &\quad \times \left. \left[1 + \frac{2\tau\rho(\bar{k} - K)}{(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)} \right] - 1 \right\}.\end{aligned}\quad (33)$$

For $\tau > \tau^*$ the squared ratio exceeds one, giving both signs. For $0 < K < \bar{k}$, the second derivative of revenue with respect to the tax is

$$\begin{aligned}R_{\tau\tau} &= -\frac{\rho(K)(1 - \tau^*)^2[\rho(K) + \rho(\bar{k} - K)]^2\rho(\bar{k} - K)}{\gamma} \\ &\quad \times \left\{ \frac{2}{[(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)]^3} + \frac{3\tau\rho(\bar{k} - K)}{[(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)]^4} \right\} < 0. \square\end{aligned}$$

If the host base is exhausted, revenue is zero and the smooth formulas no longer apply. Exhaustion occurs strictly before $\tau = 1$ only if

$$\gamma < \frac{\rho(\bar{k} - K)}{\rho(K)} + \frac{\rho(\bar{k} - K)^2}{2\rho(K)^2}. \quad (34)$$

The government therefore need not prefer a tax below confiscation at every shifting cost or investment level.

A.2 Investment Concavity

For positive profits before shifting, the mapping $-\theta^2/\rho(K)$ is jointly concave as a function of positive profit and the transfer, and increasing in profit. Hence the two after-shifting bases are concave in (K, θ) after composition with the concave profit functions $\rho(K)$ and $\rho(\bar{k} - K)$. The constraints that these bases be nonnegative therefore define a convex feasible set. If either profit is zero, only $\theta = 0$ is feasible, as established above. The weighted sum of the bases in the firm's objective is jointly concave. It is strictly concave between distinct capital choices because $1 - \tau^* > 0$ and $\rho(\bar{k} - K)$ is strictly concave. The remaining terms preserve concavity. For two different investments and their optimal transfers, a convex combination of those transfers is feasible at the intermediate investment and yields strictly more than the corresponding combination of payoffs. Maximizing over transfers therefore gives a strictly concave value function in K . On the compact investment interval its maximizer is unique. The same argument covers linear transfer limits proportional to $\rho(K)$ and $\rho(\bar{k} - K)$, whose feasible sets are convex.

A.3 Equilibrium When the Government's Type Is Known

This section establishes an equilibrium when investors know that the government is opportunistic. Fix investment at $0 < K < \bar{k}$ and suppose $0 < \tau^* < 1$. The unconstrained transfer decreases strictly with the host tax. At a low host tax, inward shifting may exhaust the foreign tax base. As the host tax rises, both reporting constraints cease to bind. A sufficiently high host tax may then exhaust the host base through outward shifting. The other two possibilities cannot be optimal. An inward transfer that exhausts the host base would reduce both countries' after-tax profits relative to zero shifting, as would an outward transfer that exhausts the foreign base.

While inward shifting exhausts the foreign base, the positive host base is fixed and revenue increases linearly with τ . When neither reporting constraint binds, Lemma 1 gives $R_{\tau\tau} < 0$ for every $\tau \geq 0$, including $\tau < \tau^*$. At the tax where the foreign base first becomes positive, the derivative of host revenue jumps downward. The firm's first-order condition gives $1 - \gamma\theta/\rho(K) > 0$, while $\theta_\tau < 0$. If outward shifting eventually exhausts the host base, revenue becomes zero. Such a tax cannot maximize revenue because a positive revenue is available at a lower rate. The government's maximizing tax is therefore unique for every $0 < K < \bar{k}$. At $K = 0$ every tax is optimal. At $K = \bar{k}$ the unique optimal tax is one. In all cases the tax best-response set is nonempty, compact, and convex.

For fixed positive γ , both payoffs are continuous on $[0, \bar{k}] \times [0, 1]$, including the endpoints under the constrained reporting technology. The firm's best response is unique by the preceding

concavity argument. Both best-response correspondences are upper hemicontinuous, and their product has nonempty compact convex values in the same set. Kakutani's fixed-point theorem therefore gives an investment and a tax that are mutual best responses for every positive γ . This establishes an equilibrium when investors know the government's type, but does not establish its uniqueness.

A.4 Equilibrium with a Common Announcement

Both types make the prescribed announcement, so investors' probability of a commitment type remains p . Strict concavity makes the solution of (10) the firm's optimal investment, taking the implemented taxes as given. The commitment type implements its announcement by assumption. The opportunistic type maximizes current revenue after any aggregate investment. Following an unexpected announcement, the specified beliefs assign probability one to the opportunistic type and firms invest K_0 . Neither type can earn more than R_0 at that investment, since R_0 is the maximum over all implemented taxes. The two announcement constraints are therefore sufficient to prevent either type from choosing another announcement. Firms choose optimal reporting after every feasible tax and investment history. Because unexpected announcements occur with probability zero in equilibrium, Bayes' rule does not determine beliefs following them. These strategies and specified beliefs constitute a weak perfect Bayesian equilibrium. Other beliefs may support other equilibria.

A.5 Proof of Proposition 1

At $p = 0$, investors know that the government will maximize revenue after investment. Investment is therefore K_0 for every nearby announcement. The derivative of the aggregate investment condition is $v_{kk} + v_{k\tau}\tau_K^O < 0$, so the implicit-function theorem gives a differentiable investment choice $K(\tau^R, p)$ near this equilibrium. Revenue when the announcement is honored is locally maximized at $\tau^R = \tau_0$. Lemma 1 establishes $R_{\tau\tau} < 0$, so this inequality need not be imposed as a separate assumption. A second application of the implicit-function theorem gives a nearby announcement $\hat{\tau}^R(p)$ satisfying the first-order condition and the strict second-order condition for a local revenue maximum.

At the reference allocation, $K_p = K_{\tau^R} = 0$, while

$$K_{\tau^R p} = -\frac{v_{k\tau}}{v_{kk} + v_{k\tau}\tau_K^O}. \quad (35)$$

Differentiating $R_\tau + R_K K_{\tau^R} = 0$, the first-order condition for revenue when the announcement is honored, gives the slope $R_K v_{k\tau} / [R_{\tau\tau}(v_{kk} + v_{k\tau}\tau_K^O)]$. Substituting $R_K = -\tau_0 v_{k\tau}$ gives (16), whose sign is negative for either nonzero sign of $v_{k\tau}$. Differentiating (10) twice along the selected announcement gives (17). Its sign is opposite to $v_{k\tau}$. Writing derivatives of the selected announcement as $\hat{\tau}^{R'}(0)$, the derivatives of selected revenue when the announcement is honored satisfy

$$\left. \frac{d\tilde{R}}{dp} \right|_0 = 0, \quad \left. \frac{d^2\tilde{R}}{dp^2} \right|_0 = -R_{\tau\tau}[\hat{\tau}^{R'}(0)]^2 > 0. \quad (36)$$

Here $\tilde{R}$ is evaluated along the one-period investment $K(\hat{\tau}^R(p), p)$. The corresponding opportunistic revenue $\hat{R}$ has first derivative zero and second derivative $-2R_{\tau\tau}[\hat{\tau}^{R'}(0)]^2 > 0$, by the envelope theorem for its implemented tax. Both revenues start at R_0 and exceed it for sufficiently small positive p , verifying the announcement constraints required for the equilibrium construction. When the announcement is honored, the tax base is $\tilde{R}/\hat{\tau}^R(p)$. Its first derivative at zero is $-R_0\hat{\tau}^{R'}(0)/\tau_0^2 > 0$.

The argument extends to any differentiable reporting technology satisfying $R = -\tau v_\tau$, provided the implemented tax is a strict optimum between zero and one with $R_{\tau\tau} < 0$. This revenue curvature must be verified for a different technology. Lemma 1 establishes it for the baseline technology when reporting constraints do not bind. Without the identity linking revenue to the firm's payoff, the slope derived before substituting that identity remains valid, and its sign depends on $R_K v_{k\tau}$. If $v_{k\tau} = 0$, the derivatives above vanish and do not establish a strict response. The proof also does not cover a zero aggregate derivative, an implemented tax of zero or one, or a point at which a reporting constraint starts to bind.

For the power profit function in the example, at $K = 1$, $\tau = \tau^*$, and $\gamma = \tau^*/[2(1 - \tau^*)]$, direct differentiation yields

$$\begin{aligned} R_\tau &= 0, & R_{\tau\tau} &= -\frac{4 - \tau^*}{2\tau^*(1 - \tau^*)}, & R_K &= \alpha\tau^*, \\ R_{\tau K} &= \alpha, & v_{k\tau} &= -\alpha, & v_{kk} &= -2\alpha(1 - \alpha)(1 - \tau^*). \end{aligned} \quad (37)$$

The tax response is $\tau_K^O = -R_{\tau K}/R_{\tau\tau} > 0$, making the aggregate investment derivative strictly negative. Substitution gives (18). The investment choice is globally optimal by strict concavity, and the implemented tax is globally optimal by the single-peaked revenue argument in Appendix A.3. The equilibrium and these strict inequalities survive sufficiently small changes in the parameters. At this reference point, $v_{k\gamma} = 0$ and $R_{\tau\gamma} > 0$. Differentiating the investment and implemented-tax first-order conditions at $p = 0$ gives

$$\frac{d\tau_0}{d\gamma} = -\frac{R_{\tau\gamma}}{R_{\tau\tau} - R_{\tau K}v_{k\tau}/v_{kk}} > 0. \quad (38)$$

A small positive cost change therefore gives $\tau_0 > \tau^*$. Continuity and $\hat{\tau}^R(0) = \tau_0$ imply outward shifting at both taxes for sufficiently small positive p in those economies.

A.6 Proof of Proposition 2

For $p > 0$ and $\tau^R < 1$, the Inada condition and strict concavity give a unique investment satisfying $0 < K < \bar{k}$. Implicit differentiation of (19) gives

$$K_{\tau^R} = -\frac{1}{(1 - \tau^R) [-\rho''(K)/\rho'(K) - \rho''(\bar{k} - K)/\rho'(\bar{k} - K)]} < 0. \quad (39)$$

The government's first-order condition is $\rho(K) + \tau^R \rho'(K) K_{\tau^R} = 0$. Substitution gives $\tau^R/(1 - \tau^R) = g(K)$ and hence (21). At any announcement satisfying the government's first-order condition, the investment condition becomes

$$p = (1 - \tau^*)[1 + g(K)] \frac{\rho'(\bar{k} - K)}{\rho'(K)}. \quad (40)$$

The right side is strictly increasing in K because g increases, the numerator marginal profit increases as foreign capital falls, and the denominator marginal profit decreases. Its limit at zero is zero because g is bounded there and $\rho'(0+) = \infty$. Its limit at $\bar{k}$ is infinity. For each $p > 0$, exactly one announcement therefore satisfies the first-order condition. Revenue is positive inside the announcement interval and vanishes at its endpoints, so this is the unique global maximum. Equation (40) gives increasing investment, and the increasing g gives an increasing announcement. At $p = 0$, investors know that the government will confiscate profits, so host investment and revenue are zero. Both types earn positive revenue at the proposed allocation for $p > 0$, so the announcement constraints hold under the specified beliefs following unexpected announcements.

For $\rho(k) = zk^\alpha$, direct substitution gives $g(K) = (1 - \alpha)\bar{k}/[\alpha(\bar{k} - K)]$, which is increasing and bounded near zero. Substitution into (21) gives (22). This calculation holds for any positive productivity z , which cancels from the tax–investment relation.

For $\rho(k) = zk^\alpha - r^*k$, the same first-order conditions give the index

$$g(K) = \frac{\rho(K)|\rho''(K)|}{\rho'(K)^2} + \frac{\rho(K)}{\rho'(K)} \frac{|\rho''(\bar{k} - K)|}{\rho'(\bar{k} - K)}, \quad (41)$$

which is (20) written using positive curvature magnitudes. All marginal profits are positive under the stated restriction. Write $u = zK^{\alpha-1}$. The first term equals $\alpha(1 - \alpha)u(u - r^*)/(\alpha u - r^*)^2$. Its derivative with respect to u has the sign of $r^*[r^* - (2 - \alpha)u]$, which is negative for $r^* > 0$. Because u decreases with K , that term increases with investment. The ratio $\rho(K)/\rho'(K)$ is strictly increasing by strict concavity. For $w = \bar{k} - K$, the logarithmic derivative of $|\rho''(w)|/\rho'(w)$ has the sign of $(2 - \alpha)r^* - \alpha zw^{\alpha-1}$, which is negative under the stronger endowment restriction. Thus the second term also increases with K , and $g'(K) > 0$. As K approaches zero, the first term converges to $(1 - \alpha)/\alpha$ and the second to zero. Thus g is bounded near zero, and all conditions of Proposition 2 hold. At $r^* = 0$ the first term is constant and the second remains strictly increasing.

A.7 Proof of Proposition 3

At the prescribed first announcement, firms assign probability π to its implementation and choose $K(\tau^R, \pi, \gamma)$. The government's gain from honoring the announcement rather than choosing its best tax revision is $\mathcal{G}$ in (25). Its sign determines the choice in (26). In particular, $0 < \sigma < 1$ requires indifference between honoring and revising the announcement, evaluated at the same installed investment and the beliefs induced by those choices. After the announcement is honored, Bayes' rule gives $q = p/[p + (1 - p)\sigma]$. Only the opportunistic type can implement a different tax, so a revision reveals its type.

Following an unexpected announcement, the specified beliefs assign probability one to the opportunistic type, inducing investment K_0 . Since investors already regard the government as opportunistic, honoring the announcement cannot improve its reputation. With no external penalty for revision, its best current revenue is R_0 . The commitment type cannot exceed that revenue at the same investment. Investors continue to regard the government as opportunistic in the second period, when both types can again obtain R_0 . The right side of (27) is consequently

each type's maximum discounted revenue from changing its first announcement. The left sides are the revenues from following the proposed strategy. These inequalities check the incentives to make the announcement, in addition to the incentives to implement it.

If aggregate investment differs from the equilibrium choice following the prescribed announcement, let the opportunistic type maximize current and discounted second-period revenue, given investors' beliefs after the tax decision. An individual firm takes aggregate investment as given and cannot change it by altering its own choice. The commitment type always implements its announcement. Second-period strategies are optimal at each specified investor belief by the one-period construction, and firms optimize reporting after every tax history. These strategies and beliefs establish sufficiency.

Conversely, sequential rationality requires optimal firm decisions and the tax-choice inequalities, while Bayes' rule determines beliefs following choices observed in equilibrium. The opportunistic type can obtain $(1 + \beta)R_0$ by making an unexpected announcement. The commitment type can attain that value by announcing τ_0 , or approach it with unexpected announcements converging to τ_0 if that rate is already the prescribed announcement. Continuity therefore makes both announcement inequalities necessary. If the two implemented taxes coincide, observing the tax conveys no information and both types can be represented as honoring the announcement. Investors cannot infer the government's type from identical tax choices.

A.8 Low Reputation without Shifting

Take any profit function satisfying Proposition 2. As $p \downarrow 0$, second-period investment and hence $R_2^O(p)$ converge to zero. When investors know that the government is opportunistic, both investment and revenue are zero. For any first-period announcement implemented with certainty, (25) requires

$$(1 - \tau^R)\rho(\tilde{K}(\tau^R)) \leq \beta R_2^O(p). \quad (42)$$

As initial reputation p approaches zero, if τ^R remains bounded away from one, the inequality forces investment and revenue to zero. If τ^R approaches one, the firm's investment when the announcement is implemented with probability one also approaches zero, by its first-order condition. Compactness of the tax interval extends this argument to every sequence of feasible announcements. First-period revenue therefore vanishes throughout this family of equilibria in which both types honor the announcement. This result concerns a two-period horizon. With an infinite horizon, the loss of future revenue after a tax revision need not vanish as p approaches zero, so this argument does not establish the same conclusion.

B Extensions and Computation of the Reputation Model

This appendix develops the qualifications to the reputation mechanism discussed in Sections 4.1 and 5.1, then documents the numerical results in Sections 4.1.4 and 4.2.2. Appendix B.1 extends the local tax result to heterogeneous firms, and Appendix B.2 examines how the mobile share

of the base and the available tax instruments affect the incentive to revise a tax promise. Appendix B.3 supplies the cost formulas and adjustment-limit comparisons underlying the numerical sensitivity analysis. Appendices B.4 and B.5 describe the solution checks, report the allocations behind the figures, and vary the value of future reputation.

B.1 Heterogeneous Investment and Reporting Responses

Section 5.3 asks whether the tax result requires identical firms. This extension shows that the local decline in Proposition 1 also holds with heterogeneous investment responses under the aggregate condition below. It also separates the tax conclusion from the stronger condition needed for aggregate investment to rise. Let group i have population weight $w_i > 0$, optimized investment payoff $v_i(k_i, \tau)$, and its own profit function and shifting cost. Under a common tax, aggregate revenue is $R = -\tau \sum_i w_i v_{i,\tau}$. At an equilibrium where investors know that the government is opportunistic and $0 < \tau_0 < 1$ is a strict optimum with differentiable revenue, define $k'_i(\tau) = -v_{i,k\tau}/v_{i,kk}$, allowing different signs across groups and requiring $v_{i,kk} < 0$. The condition on the interaction between investment and the implemented tax is

$$1 - \sum_j \tau_{K_j}^O k'_j(\tau_0) > 0. \quad (43)$$

No sign is imposed on any individual tax response $\tau_{K_j}^O$. If no reporting constraint binds in a neighborhood of the equilibrium, Lemma 1 gives negative tax curvature for every shifting group. Immobile groups contribute linear revenue. Thus, aggregate $R_{\tau\tau} < 0$ follows whenever at least one positive-weight group uses the baseline shifting technology with nonbinding reporting constraints.

Differentiating each investment condition under uncertainty about the implemented tax at $p = 0$ gives

$$K_{i,\tau^R p} \Big|_0 = \frac{k'_i(\tau_0)}{1 - \sum_j \tau_{K_j}^O k'_j(\tau_0)}. \quad (44)$$

Indeed, $K_{i,\tau^R p} = k'_i(1 + \sum_j \tau_{K_j}^O K_{j,\tau^R p})$. Multiplication by $\tau_{K_i}^O$ and summation solve for the common factor. Differentiating the announcement first-order condition and using $R_{K_i} = -\tau_0 w_i v_{i,k\tau}$ then gives

$$\begin{aligned} \frac{d\hat{\tau}^R}{dp} \Big|_0 &= -\frac{\sum_i R_{K_i} k'_i(\tau_0)}{R_{\tau\tau} [1 - \sum_j \tau_{K_j}^O k'_j(\tau_0)]} \\ &= -\frac{\tau_0 \sum_i w_i (v_{i,k\tau})^2 / v_{i,kk}}{R_{\tau\tau} [1 - \sum_j \tau_{K_j}^O k'_j(\tau_0)]} < 0, \end{aligned} \quad (45)$$

provided at least one group has $v_{i,k\tau} \neq 0$. The decline therefore does not require every group's investment to respond negatively to tax, or the opportunistic government's implemented tax to increase with every group's capital. Each group's first investment derivative along the selected announcement is zero, and its second derivative is $2K_{i,\tau^R p} \hat{\tau}^{R'}(0)$. Investment rises in groups with $k'_i < 0$ and falls in groups with $k'_i > 0$. Aggregate investment rises if $\sum_i w_i k'_i < 0$.

The second derivatives of revenue with respect to p at zero, for the commitment and opportunistic types respectively, remain $-R_{\tau\tau} [\hat{\tau}^{R'}(0)]^2$ and $-2R_{\tau\tau} [\hat{\tau}^{R'}(0)]^2$. They verify both

announcement constraints as in Appendix A.5. The tax base when the announcement is honored also rises. For other reporting technologies, the same result holds when the identity linking revenue to the firm's payoff and strict tax curvature hold, but curvature must then be verified separately. The proof does not cover confiscation, a point where a reporting constraint starts to bind, or a discontinuous change between revenue-maximizing taxes. An average local reporting elasticity does not exclude those cases.

B.2 Tax Instruments and Profit Mobility

The heterogeneity result requires the tax chosen by a government known to be opportunistic to lie strictly between zero and one. As discussed in Section 5.3, firms with mobile profits need not make such a tax globally optimal when much of the base is immobile. I first compare a common tax with separate group-specific taxes, then derive the mobile share required for a common tax below one to yield more revenue than confiscation. For firms with sunk investments K_i , permitting separate implemented taxes weakly increases the current gain from revising a common announcement. In particular,

$$\sum_i \max_{\tau_i} R_i(\tau_i, K_i) - \sum_i R_i(\tau^R, K_i) \geq \max_{\tau} \sum_i R_i(\tau, K_i) - \sum_i R_i(\tau^R, K_i). \quad (46)$$

The feasible set of tax vectors contains every uniform vector. A separately taxed immobile group with a positive base is taxed at one when the government maximizes revenue after investment. These statements concern current payoffs at fixed investments. Different tax instruments can also change second-period investment and taxation, so these current revenue comparisons do not determine the effects over two periods.

Even under a common tax, a small mobile group need not prevent confiscation. Suppose a share w has per-firm revenue $R_m(\tau)$, with m denoting mobile profits, with $R_m(1) = 0$, and the remaining share has immobile base $\rho(k_n) > 0$, with n denoting the group without mobile profits. A tax $\tau < 1$ yields more than confiscation precisely when

$$wR_m(\tau) > (1 - w)(1 - \tau)\rho(k_n). \quad (47)$$

The threshold mobile share is therefore

$$w_c = \inf_{\tau < 1: R_m(\tau) > 0} \frac{(1 - \tau)\rho(k_n)}{(1 - \tau)\rho(k_n) + R_m(\tau)}. \quad (48)$$

For $w > w_c$, some tax below one yields strictly more revenue than confiscation. For $w < w_c$, no such tax does. At $w = w_c$, equality of revenues depends on whether a tax attains the infimum in (48). If the mobile base is exhausted below one, the threshold need not be zero. This identifies when the tax below confiscation required by the local reputation result is preferred to confiscation.

The empirical concentration documented by Wier and Erasmus (2023) makes the size of the mobile base relevant, but a group's share of firm numbers need not equal its share of the tax base. With equal profits before shifting and reference semi-elasticity 1.26, the threshold in equation (48) is approximately 0.712 for deductible costs in both countries, 0.577 without deductibility, and 0.750 for deductible costs at the origin. These hold investment fixed and are not an estimated distribution of firms or a heterogeneous dynamic equilibrium. They show why

matching an average local elasticity in Table 4 cannot by itself establish that the government prefers a tax below confiscation.

External Enforcement. The government's incentive constraint in Section 4.2 also allows a direct comparison of current revision incentives when an external penalty is added. External enforcement can strengthen the incentive to honor the announcement. A first-period penalty χ for revision adds χ to the gain from honoring the announcement in (25), holding investment and second-period revenues fixed. It thus weakly expands the set of promises that both types are willing to honor at those payoffs. Determining how the selected announcement, investment, or the probability of honoring the announcement changes requires solving investment, beliefs, and the payoff from an unexpected announcement again under the penalty. Compensation paid to firms additionally changes their investment payoffs. These distinctions connect the analysis to Kohler and Stähler (2019) and Horn and Tangerås (2021). Treaty counts are neither an exogenous penalty measure nor a direct test of the comparative static.

B.3 Cost Treatment and Adjustment Limits

This section supplies the reporting rules for Table 4 and the transfer-limit exercise in Section 5.2. The two exercises address different assumptions. Changing cost treatment holds the response to a small tax change fixed at the reference allocation. A transfer limit instead changes how much profit remains adjustable after a large tax revision. With non-deductible bilateral resource costs, the firm's objective is

$$(1 - \tau)[\rho(K) + \theta] + (1 - \tau^*)[\rho(\bar{k} - K) - \theta] - \frac{\gamma\theta^2}{2} \left[\frac{1}{\rho(K)} + \frac{1}{\rho(\bar{k} - K)} \right]. \quad (49)$$

The unconstrained transfer is $(\tau^* - \tau)\rho(K)\rho(\bar{k} - K)/[\gamma(\rho(K) + \rho(\bar{k} - K))]$, subject to nonnegative bases. When neither reporting constraint binds, the revenue first-order condition gives the tax $\tau^*/2 + \gamma[\rho(K) + \rho(\bar{k} - K)]/[2\rho(\bar{k} - K)]$, which can exceed one. The government must therefore also compare revenue at taxes of zero and one and at taxes where a reporting constraint starts to bind.

If deductible costs arise only where profit originates, an outward transfer $x = -\theta \geq 0$ gives bases $\rho(K) - x - \gamma x^2/[2\rho(K)]$ and $\rho(\bar{k} - K) + x$. The unconstrained outward transfer is $x = (\tau - \tau^*)\rho(K)/[\gamma(1 - \tau)]$. Host revenue then scales with $\rho(K)$, so the revenue-maximizing tax is independent of K while the reporting constraints remain nonbinding. The local reputation result can still apply with $\tau_K^O = 0$, but a channel operating through τ_K^O is absent. Inward shifting assigns costs to the foreign origin and is solved separately.

Adjustment limits restrict the transfer to $-m\rho(K) \leq \theta \leq m\rho(\bar{k} - K)$, intersected with the baseline feasible interval, where $0 \leq m \leq 1$. The parameter describes the fraction of current profits that existing arrangements can reallocate after policy becomes known. At $m = 0$ no adjustment is possible. A separate fixed participation cost would require comparing zero shifting with the best nonzero transfer net of that cost. A derivative calculation then describes

the response only over tax ranges in which firms continue to make the same participation choice.

For Table 4, I choose the cost parameter to give a common tax-base semi-elasticity ε at equal taxes and equal profits before shifting, holding investment fixed. The implied parameters are $\gamma = 1/[2\varepsilon(1 - \tau^*)]$ when costs arise in both countries and are deductible, $\gamma = 1/(2\varepsilon)$ when they arise in both countries and are not deductible, and $\gamma = 1/[\varepsilon(1 - \tau^*)]$ when deductible costs arise only in the country from which profits are shifted. Equal semi-elasticities at this point do not imply equal responses at every equilibrium. Table 5 varies m at $\varepsilon = 1.26$ with deductible costs in both countries. Nonnegative tax bases restrict transfers even in the row with no additional limit, where $m = 1$.

The decomposition in (15) explains the declining tax in the cost comparisons. With deductible costs in both countries, $\varepsilon = 1.26$, $m = 1$, and $p = 0.5$, the term for investment expansion is approximately 0.076 and the term measuring the change in investment responsiveness is -0.841 . The latter dominates. The replication file reports both components for all eighteen allocations in Table 4.

TABLE 5: Announced taxes when profit transfers are limited

Maximum share shifted	Announced tax rate (%)		
	$p = 0.1$	$p = 0.5$	$p = 0.9$
No additional limit	31.5	29.3	27.2
5%	50.0	40.9	44.5
15%	87.7	49.4	49.4
60%	100.0	100.0	26.7

Notes: Announced rates are in percent. All rows use deductible costs in both countries and reference semi-elasticity 1.26. The limit is a share of profits earned in the country from which they are shifted. The unrestricted row repeats the corresponding cost comparison in Table 4. Nonnegative tax bases apply throughout.

TABLE 6: Additional one-period comparisons of shifting costs and transfer limits

Cost specification	γ	Announced tax rate (%)		
		$p = 0.1$	$p = 0.5$	$p = 0.9$
Both countries, deductible costs	8	66.0	52.0	43.3
Both countries, deductible costs	20	72.5	51.7	43.9
Both countries, deductible costs	100	78.8	47.7	42.3
Both countries, deductible costs	1000	67.5	37.3	42.4
Both countries, non-deductible costs	1.25	73.3	49.1	39.7
Deductible costs where profits originate	1.25	34.3	32.6	30.8
Baseline costs, 15% transfer limit	1.25	88.5	49.6	49.5
Baseline costs, 60% transfer limit	1.25	45.2	40.3	35.9

Notes: Announced taxes are in percent. All cases use the benchmark production parameters. The transfer-limit rows vary m with deductible costs in both countries. Unlike Table 4, the reference elasticity is not matched across cost specifications. Very high costs examine a boundary where profits are less mobile and are not an empirical calibration range.

B.4 Numerical Solution and Verification

The figures in Sections 4.1.4 and 4.2.2 require firms and governments to prefer their reported choices. The checks below compare payoffs from alternative choices in addition to evaluating equation residuals. They verify incentives at the computed equilibria without establishing that no other equilibrium exists. The one-period solver finds investment fixed points with firms taking the implemented taxes as given, brackets candidate announcement maxima, and refines their locations continuously. It independently checks firms' global investment payoffs, the government's implemented-tax payoff, and both announcement constraints. At each reported baseline optimum, an independent 201-point investment scan finds one sign crossing. This diagnostic is not a proof of global equilibrium uniqueness. Increasing the announcement scan from 65 to 129 points changes the central tax by less than 3.8×10^{-6} , investment by less than 1.3×10^{-5} , and revenue when the announcement is honored by less than 8×10^{-12} .

The two-period computation uses $q \in [p, 1]$, the probability investors assign to the commitment type after the announcement is honored. This probability implies $\pi = p/q$ and $\sigma = (p/q - p)/(1 - p)$. For each q strictly between p and one, the computation finds announcements at which the opportunistic type is indifferent between honoring and revising, then checks both types' incentives to make that announcement. At $q = p$ and $q = 1$, it also considers the inequalities allowing the government always to honor or always to revise. The search covers these cases and refines candidate maxima between them. This refinement allows occasional revision even when the probability of honoring is close to one. The implemented-tax maximization includes zero and one. Omitting a tax of one would miss confiscation when limits on profit transfers leave a positive host tax base.

The two-period comparisons contain thirty computed equilibria, combining five initial reputation levels, three discount factors, and two reporting technologies. For six comparisons at $\beta = 0.6$, doubling the grids for q , announcements, and second-period payoffs changes the announcement by at most 6.4×10^{-5} , σ by at most 5.1×10^{-5} , and investment by at most 1.4×10^{-5} . The final verification computes second-period optimal choices directly in place of interpolated values and refines each solution requiring indifference between honoring and revising. Across all thirty equilibria, the largest gain from changing the firm's investment is below 9×10^{-16} , the largest gain from changing the tax chosen after revision is below 3×10^{-16} , and the error in an implementation-indifference condition is below 2.6×10^{-11} . Both types strictly prefer the prescribed announcement to their best alternative announcement under the specified beliefs. These checks verify the reported incentives to numerical tolerance. They do not establish global uniqueness or select beliefs beyond the equilibrium definition.

The local reputation theorem is also checked independently. Symbolic differentiation recovers the derivatives for the power profit function in Appendix A.5 and the second derivatives of both types' revenues. At $\alpha = 0.66$, $\tau^* = 0.15$, $\bar{k} = 2$, zero rental deduction, and $\gamma = \tau^*/[2(1 - \tau^*)]$, the analytical tax slope at zero is -0.010547 . The separately solved finite difference between initial reputation levels 0.01 and 0.02 is -0.010488 . This calculation checks the implementation of a proven local result. It does not replace the proof.

B.5 Additional Numerical Results

Tables 7 and 9 provide the exact allocations underlying the quantitative discussion. Figure 1 in the main text displays the baseline static mechanism, and Figure 2 shows tax choices at the central discount factor. Figure 3 in Section 5.4 compares all three discount factors. Each point comes from a computed equilibrium in which both types make the same announcement. Lines connect the computed points. Monetary quantities are in model units. Table 8 reports all 27

TABLE 7: Reputation and the one-period tax announcement

γ	p	Announced tax τ^R	Host investment	Tax base	Revenue
0.50	0.1	0.324	3.549	1.524	0.494
0.50	0.5	0.300	3.651	1.664	0.500
0.50	0.9	0.279	3.909	1.848	0.515
1.25	0.1	0.452	2.442	1.210	0.547
1.25	0.5	0.403	2.632	1.397	0.563
1.25	0.9	0.359	3.153	1.693	0.607
4.00	0.1	0.595	1.246	0.852	0.507
4.00	0.5	0.497	1.552	1.105	0.549
4.00	0.9	0.419	2.472	1.587	0.665

Notes: All rows use $\alpha = 0.66$. Investment, the tax base when the announcement is honored, and revenue are in model units. Each row checks firm and implemented-tax best responses and both types' announcement incentives under the specified investor beliefs. The complete outputs also vary α over 0.5, 0.66, and 0.8.

saved one-period comparisons across production curvature, shifting costs, and reputation. It complements the cost-treatment exercise in Section 5.1 by changing the curvature of investment returns within the baseline reporting technology.

TABLE 8: One-period announcements across production curvature and shifting costs

Production curvature α	Shifting cost γ	Announced tax rate (%)		
		$p = 0.1$	$p = 0.5$	$p = 0.9$
0.50	0.50	34.6	33.2	31.9
0.50	1.25	50.1	46.8	44.0
0.50	4.00	67.4	60.4	54.9
0.66	0.50	32.4	30.0	27.9
0.66	1.25	45.2	40.3	35.9
0.66	4.00	59.5	49.7	41.9
0.80	0.50	29.6	25.8	22.4
0.80	1.25	39.9	32.3	26.3
0.80	4.00	51.9	35.8	28.3

Notes: Announced rates are in percent. The profit function is $\rho(k) = k^\alpha - 0.04k$, with $\bar{k} = 10$ and $\tau^* = 0.15$. Shifting costs are deductible in both countries. Each row varies reputation while holding the two displayed parameters fixed.

TABLE 9: Tax policies and investment in the two-period model

Reporting technology	p	τ^R	σ	q	Investment	Revision (pp)
Profit shifting	0.1	0.454	1.000	0.100	2.510	0.00
Profit shifting	0.3	0.432	1.000	0.300	2.673	0.00
Profit shifting	0.5	0.409	1.000	0.500	2.854	0.00
Profit shifting	0.7	0.384	1.000	0.700	3.054	0.00
Profit shifting	0.9	0.358	1.000	0.900	3.272	0.00
No shifting	0.1	0.550	0.345	0.244	0.153	26.54
No shifting	0.3	0.594	0.670	0.390	0.633	9.39
No shifting	0.5	0.598	1.000	0.500	1.191	0.00
No shifting	0.7	0.513	1.000	0.700	1.854	0.00
No shifting	0.9	0.448	1.000	0.900	2.416	0.00

Notes: The two-period calculation allows the opportunistic type always to revise ($\sigma = 0$), sometimes to revise ($0 < \sigma < 1$), or always to honor ($\sigma = 1$) its announcement. The probability investors assign to commitment after the promise is honored is q . Both economies use the benchmark production parameters and $\beta = 0.6$. Shifting costs are $\gamma = 1.25$. The expected tax revision is $100(1 - \pi)[\tau^O(K, \gamma) - \tau^R]$, in percentage points. It is not an accounting effective-tax-rate gap.

C Data and Additional Empirical Results

This appendix supports the three empirical comparisons in Section 2. Appendix C.1 documents the ICRG reconstruction, and Appendix C.2 assesses the relationship between risk and taxes in Section 2.2. Appendix C.3 examines the persistence of contract risk and alternative observation intervals. Appendix C.4 gives the complete FDI and profit shifting comparisons behind Sections 2.3 and 2.4. Appendices C.5 and C.6 construct and check the shares of profits and employment used in Section 2.4. Appendix C.7 explains the distinction between these observations and the government’s tax decision in Section 4.2.

C.1 ICRG Sources and Annual Construction

The contract risk variable defined in Section 2.1 is reconstructed from monthly observations before it enters any of the main regressions. This section explains the calendar-year and country matching choices. The monthly ICRG workbook separates Contract Viability/Expropriation, Profits Repatriation, Payment Delays, and their aggregate Investment Profile. Each component is scored from zero to four, and the aggregate from zero to twelve, with higher scores denoting lower risk. The primary variable is four minus Contract Viability/Expropriation. The alternative aggregate measure is twelve minus Investment Profile. These measures have different units and economic coverage, so their coefficients are not directly comparable in magnitude. Country identifiers are matched by name and documented aliases. Historical Serbia–Montenegro is kept separate from Serbia.

The baseline averages all twelve monthly observations in a calendar year. It therefore omits the partial 2001 coverage and other incomplete country-years. The extracted monthly file of contract scores agrees exactly with the provider workbook. The inherited merged annual contract risk series cannot be reproduced by calendar-year averaging, values in the first or last month, or a shift of one year. The replication archive retains both series and a comparison of every observation. All main estimates use the source reconstruction rather than that inherited annual series. A separate comparison includes partial-year averages, without imputing missing months.

The tax-specific panel is matched to the archived workbook of statutory tax rates and the World Bank series for real GDP per capita in constant 2015 US dollars, population, and total natural resource rents as a share of GDP. Statutory rates agree with their archived source. Each risk measure uses a common sample with all variables observed across controls. The country means average those same country-years for the rate, risk, and controls. Annual lags are matched by country and calendar year, so a gap in observations cannot become a lag of one year. The replication files record the country-year panel and the country inventory for every specification. Restricting only to the covariates needed for the model with linear income controls produces the same baseline sample of country averages.

C.2 Robustness of the Relationship between Risk and Taxes

Section 2.2 reports the within-country association conditional on income, population, and resource rents. Here I compare it with the relationship across countries and examine which time variation supports it. The samples and additional specifications address persistence and country composition rather than supply independent tests of a causal reputation effect. The comparison of country means separates the within-country result from persistent cross-country differences. For country h , I estimate

$$\bar{\tau}_h^{\text{stat}} = \alpha + \beta \bar{q}_h + \delta_y \overline{\log y_h} + \delta' \bar{Z}_h + \eta_{r(h)} + u_h, \quad (50)$$

where bars average jointly observed country-years and $\eta_{r(h)}$ denotes continent effects. The unadjusted regression omits all controls. The comparison controlling for income adds only average log income, while the full comparison also includes log population, resource rents, and continent effects. Standard errors are HC3. On the common 135-country sample, the raw coefficient on contract risk is 3.259 (SE 1.092). Controlling for income reduces it to 0.530 (SE 1.417), and the full linear controls yield 0.880 (SE 1.392). Thus the cross-country comparison alone does not establish a relationship independent of development.

Table 10 reports the primary risk coefficient under the three control sets in the main table. In addition to the full annual sample, I use 2010–2019, countries observed in every year from 2003 to 2021, observations with positive tax rates, and countries outside the original haven classification. The latter is a broad inherited classification, not a claim that every included jurisdiction has the same tax regime. A balanced country must meet the common observation requirements in all nineteen years.

For the trend comparison, I add $\xi_h t$ to equation (1). The first-difference equation is

$$\Delta \tau_{ht}^{\text{stat}} = \Delta \lambda_t + \beta \Delta q_{h,t-1} + \delta_y \Delta \log y_{ht} + \delta' \Delta Z_{ht} + \Delta u_{ht}. \quad (51)$$

Differences require adjacent calendar years. Period averages use four consecutive blocks, 2003–2007, 2008–2012, 2013–2017, and 2018–2021. For block b , I estimate

$$\bar{\tau}_{hb}^{\text{stat}} = \alpha_h + \lambda_b + \beta \bar{q}_{h,b}^L + \delta_y \overline{\log y_{hb}} + \delta' \bar{Z}_{hb} + u_{hb}, \quad (52)$$

where $\bar{q}_{h,b}^L$ averages the lagged annual scores used in equation (1) over observed years in block b . The period comparison contains 513 country-block observations. It reduces annual variation but retains the same chronological alignment of scores and taxes.

The coefficient on contract risk is positive in all 24 comparisons in Table 10. Its precision varies. The 2010–2019 and comparisons of countries observed in every year retain positive estimates under every displayed control set. Excluding the original havens gives smaller and less precise estimates. The coefficients after controlling for trends are between 0.326 and 0.342 and are not distinguishable from zero at conventional levels. First differences give approximately 0.26, while period averages give estimates between 0.91 and 1.13. These comparisons do not represent independent replications of the same test. They show how the result changes when persistent differences, national trends, and annual variation contribute differently to the estimate.

TABLE 10: Contract risk and taxes within countries

	(1)	(2)	(3)
Log income per capita	Yes	Yes	Yes
Log population	No	Yes	Yes
Resource rents	No	No	Yes
Full annual sample	0.781** (0.336)	0.664** (0.331)	0.641* (0.335)
2010–2019	0.658** (0.262)	0.695** (0.270)	0.693** (0.272)
Balanced countries	0.949** (0.377)	0.904** (0.355)	0.889** (0.364)
Positive tax rates	0.761** (0.343)	0.624* (0.341)	0.605* (0.344)
Excluding original havens	0.588* (0.345)	0.550 (0.356)	0.527 (0.357)
Country-specific trends	0.342 (0.347)	0.326 (0.350)	0.326 (0.351)
First differences	0.258* (0.154)	0.266* (0.151)	0.265* (0.152)
Period averages	1.126** (0.534)	0.969* (0.518)	0.909* (0.530)

Notes: Coefficients on contract risk in the preceding calendar year, with country-clustered standard errors. Level and period-average regressions include country and time effects. First differences include year effects. All controls in a first-difference specification are differenced. Period averages require at least three observations per block. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Omitting each country from the full annual sample gives a coefficient range of [0.686, 0.892] with linear income. Every individual omission excludes zero using pointwise 95% intervals. These exclusions assess whether one country drives the baseline result. They do not eliminate its sensitivity to separate country trends. Clustered covariance estimates use the finite-sample factor $G/(G-1)(N-1)/(N-k)$, where G is the number of countries and k the rank of the regression design. Inference uses $G-1$ degrees of freedom. Exact redundancies among country trends and year effects are removed before computing this covariance.

Including partial calendar years preserves the main direction. The aggregate Investment Profile measure, however, gives weaker annual associations. This difference matters because repatriation restrictions and payment delays need not have the same connection to statutory taxation as contract protection. Regressions of country means with income controls are imprecise for both measures. The complete set of seven control specifications, five samples, and the comparisons using the inherited series is available in the replication files. The sensitivity analysis using variation within countries was specified after observing the reconstructed baseline result, so its intervals are interpreted as descriptive, pointwise inference. Public Government Effectiveness and Rule of Law comparisons remain supplementary diagnostics and do not replace the primary ICRG measure.

TABLE 11: Annual coverage and alternative ICRG measures

	Country means		Annual country and year FE	
	Yes	Yes	Yes	Yes
Log income per capita	No	Yes	No	Yes
Population and rents				
Contract risk, full years	0.530 (1.417)	0.880 (1.392)	0.781** (0.336)	0.641* (0.335)
Observations	135	135	2424	2424
Contract risk, partial years included	0.775 (1.447)	1.164 (1.442)	0.731** (0.325)	0.596* (0.329)
Observations	136	136	2548	2548
Aggregate investment risk	0.169 (0.461)	0.536 (0.489)	0.161 (0.140)	0.107 (0.143)
Observations	135	135	2424	2424

Notes: Contract risk ranges from zero to four, and aggregate investment risk from zero to twelve. Models of country means use HC3 standard errors. Annual models use lagged risk and country-clustered standard errors. Full controls include log population, resource rents, and linear income, with continent effects in the models of country means. Partial-year coverage retains observed monthly means. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.3 Persistence and the Observation Interval

Facts 1 and 2 use changes within countries and differences across countries, respectively. Table 12 measures the variation available for each comparison. In the tax sample, adjacent annual contract risk scores have correlation 0.937, and 59.0% of adjacent pairs have identical scores. Differences in country means account for 69.4% of the total sum of squared deviations in risk. The standard deviation falls from 0.678 to 0.375 after removing country means, and to 0.330 after removing country and year effects and the three controls. The FDI sample exhibits almost the same persistence and residual variation.

TABLE 12: Persistence and variation in contract risk

	Statutory tax rate		FDI/GDP	
	All	No havens	All	No havens
Adjacent-year correlation	0.937	0.933	0.937	0.933
Unchanged scores (%)	59.0	57.2	59.3	57.6
Variance between countries (%)	69.4	67.7	69.5	67.6
Standard deviation	0.678	0.674	0.682	0.677
After country effects	0.375	0.384	0.377	0.386
After country and year effects	0.337	0.343	0.339	0.346
After effects and controls	0.330	0.337	0.330	0.337
Country-year observations	2,424	2,050	2,552	2,172
Countries	135	115	136	116

Notes: The variable is the preceding calendar year's contract risk in each estimation sample. Correlations and unchanged shares use adjacent observed calendar years. The share between countries weights country means by their observation counts. Standard deviations use the number of observations as the denominator. Controls are log real income, log population, and resource rents.

Country effects therefore remove much of the variation in risk, but considerable variation remains. Its statistical content depends on its covariance with the outcome and the uncertainty of that covariance. Persistence alone does not establish that a weak estimate reflects low power. If the score measures an underlying risk with error, transformations that remove country effects

can magnify the resulting attenuation (Griliches and Hausman, 1986). Neither the amount nor the time dependence of measurement error is identified here. I consequently compare observation intervals without interpreting them as a correction for measurement error or using lagged scores as instruments.

Table 13 applies the same controls and observation rules to taxes and FDI. All regressions include log real income, log population, and resource rents. Country means average the jointly observed years in each outcome sample. Annual and period-average regressions include country and time effects. Period averages use the four blocks defined in equation (52) and require at least three observed years per block. The final block spans four years rather than five. For differences over ℓ years, I estimate

$$\Delta_\ell Y_{ht} = \lambda_t^\ell + \beta_\ell \Delta_\ell q_{h,t-1} + \delta'_\ell \Delta_\ell X_{ht} + e_{ht}^\ell, \quad \ell \in \{1, 3, 5\}, \quad (53)$$

where Y is the statutory rate or net FDI/GDP, $\Delta_\ell Y_{ht} = Y_{ht} - Y_{h,t-\ell}$, and X contains the same three controls. Each difference requires both exact calendar endpoints. Country effects cancel, and endpoint year effects absorb common changes. Standard errors cluster by country, retaining the dependence among overlapping differences. These are relationships among changes over the stated intervals, not cumulative responses to an identified risk shock.

TABLE 13: Contract risk, taxes, and FDI across observation intervals

	Statutory tax rate		FDI/GDP	
	All	No havens	All	No havens
Country means	0.520 (1.442)	-0.339 (1.378)	-3.056* (1.597)	-1.424** (0.610)
Observations	135	115	136	116
Annual observations	0.641* (0.335)	0.527 (0.357)	0.027 (1.484)	0.250 (0.400)
Observations	2,424	2,050	2,552	2,172
Period averages	0.909* (0.530)	0.756 (0.563)	-0.862 (1.595)	-0.111 (0.577)
Observations	513	434	537	457
One-year differences	0.265* (0.152)	0.224 (0.170)	1.979 (1.397)	1.112* (0.619)
Observations	2,285	1,932	2,416	2,056
Three-year differences	0.383 (0.256)	0.356 (0.290)	0.990 (1.081)	0.632 (0.400)
Observations	2,010	1,698	2,144	1,824
Five-year differences	0.580* (0.319)	0.497 (0.351)	1.003 (0.905)	0.398 (0.575)
Observations	1,742	1,469	1,872	1,592

Notes: Each entry is the coefficient on risk in a separate regression. Both outcomes are in percentage points. Country means use HC3 standard errors. Other regressions cluster by country. All columns control for income, population, and resource rents. FDI retains zero and negative flows. Stars indicate pointwise inference. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The tax coefficient remains positive in every comparison using time variation. In the full sample, its magnitude is 0.641 annually, 0.909 in period averages, and 0.580 in five-year differences. Their respective standard errors are 0.335, 0.530, and 0.319. Longer intervals therefore do not deliver uniformly greater precision. Excluding havens further weakens the estimates. For FDI, country means retain the negative association reported in Fact 2, while

the within-country estimates remain imprecise and can have a different sign. Averaging over several years does not turn that cross-country pattern into evidence of the same response within a country.

These comparisons follow inspection of the earlier annual and country-mean estimates and are descriptive sensitivity analysis. Within each outcome and sample, adjusting the three difference tests by Holm's method gives no rejection at the 5% level. The replication files retain all 24 coefficients, confidence intervals, adjusted difference-test probabilities, and estimation samples. Persistent differences across investment destinations therefore need not describe the response to a change in contract risk within a destination.

C.4 Additional FDI and Profit-Shifting Regressions

Section 2.3 reports the relationship between contract risk and average FDI, and Section 2.4 separates reported profits from real activity. The additional comparisons here show the effect of the controls, haven exclusions, and the original log-flow outcome. They use the reconstructed ICRG scores throughout.

FDI Relative to GDP. For comparison with country averages, the annual regression is

$$f_{ht} = \alpha_h + \lambda_t + \beta q_{h,t-1} + \delta' X_{ht} + u_{ht}. \quad (54)$$

Here f is net FDI/GDP, q is contract risk, and X contains log real income, log population, and resource rents. Table 14 reports all eight comparisons from equations (2) and (54). Raw and controlled estimates use the same sample with all variables observed. The FDI data contain 2,552 country-years, including 159 negative and seven zero net inflows. Both signs are retained. The relationship in country averages remains negative with controls in both host samples. Annual estimates with country and year effects are imprecise and close to zero. The discrepancy is economically relevant because persistent differences in destinations need not describe the response to a change in risk within a destination.

TABLE 14: FDI regressions with and without income controls and havens

	(1)	(2)	(3)	(4)
<i>Country averages</i>				
Contract risk	-3.995** (1.970)	-3.056* (1.597)	-0.114 (0.400)	-1.424** (0.610)
Observations	136	136	116	116
Countries	136	136	116	116
<i>Annual observations with country and year effects</i>				
Contract risk	0.274 (1.418)	0.027 (1.484)	0.285 (0.382)	0.250 (0.400)
Observations	2552	2552	2172	2172
Countries	136	136	116	116
Income, population and rents	No	Yes	No	Yes
Exclude original havens	No	No	Yes	Yes

Notes: FDI is net inward investment as a percentage of GDP. Risk is lagged one calendar year. Country means use HC3 standard errors and annual regressions cluster by country. The raw and controlled columns use the same observations within each host sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Profit Shifting and the Control for Statutory Taxes. For country h in 2016, I estimate

$$PS_h = \alpha + \beta q_h^j + \delta_\tau \tau_h^{\text{stat}} + \delta' W_h + \zeta' X_h + u_h, \quad (55)$$

The dependent variable PS is estimated net inward profit shifting as a percentage of GDP. A negative value denotes net outward shifting. The risk measure q^j is either contract risk or aggregate investment risk, entering a separate regression. The original controls W are the index of capital controls on direct investment of [Fernández et al. \(2016\)](#), real GDP growth, and the manufacturing share. The additional controls X are log income, log population, and resource rents. These are estimated reallocations of profits, not observed transactions. Table 15 reports all twelve cross-sectional comparisons underlying equation (55). In the 116-country sample, the coefficient on contract risk changes from -2.432 to -2.325 when the statutory rate is added. The coefficient on aggregate investment risk changes from -0.984 to -0.950 . Income, population, and resource rents explain substantially more of these associations than the statutory rate alone. Excluding havens changes the host population and weakens the original relationship. The two risk variables enter separately and have different scales.

TABLE 15: Profit shifting regressions with nested controls

	(1)	(2)	(3)
<i>All countries</i>			
Contract risk	-2.432** (1.098)	-2.325** (1.159)	-0.483 (0.962)
Investment risk	-0.984*** (0.360)	-0.950** (0.375)	-0.376 (0.422)
Countries	116	116	116
<i>Excluding original havens</i>			
Contract risk	-0.128 (0.573)	-0.090 (0.569)	0.798 (0.787)
Investment risk	-0.170 (0.209)	-0.155 (0.207)	0.133 (0.321)
Countries	99	99	99
Original controls	Yes	Yes	Yes
Statutory tax rate	No	Yes	Yes
Income, population and rents	No	No	Yes

Notes: The dependent variable is estimated net inward profit shifting as a percentage of GDP in 2016. Each risk coefficient comes from a separate regression. Original controls are capital controls, GDP growth, and the manufacturing share. HC3 standard errors condition on the published outcome estimates. Samples are held fixed across columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The Original Outcome Definitions and Samples. I also apply the source reconstruction to the original FDI and profit-misalignment specifications. This separates a change in the risk measure from a change in the regression design. Table 16 reports the investment comparison. The annual log specification is

$$\log F_{ht} = \alpha_h + \lambda_t + \beta q_{h,t-1} + \varphi \log F_{h,t-1} + u_{ht}, \quad (56)$$

where F denotes net inward FDI and the logarithm requires positive current and lagged flows. Column (1) omits country effects. Columns (3) and (4) instead regress average net FDI/GDP by country on average lagged risk without other covariates. These ratios retain zero and negative net flows. They are quantities from the balance of payments rather than the model's installed productive capital.

TABLE 16: Contract risk and the original FDI comparisons

	Log net FDI		Mean net FDI/GDP	
	(1)	(2)	(3)	(4)
Lagged contract risk	-0.113*** (0.033)	-0.089 (0.057)	-3.888** (1.838)	-0.216 (0.371)
Lagged dependent variable	Yes	Yes	No	No
Country effects	No	Yes	No	No
Year effects	Yes	Yes	No	No
Exclude original havens	No	No	No	Yes
Observations	2309	2309	137	117

Notes: The ICRG measure is reconstructed from full calendar years. Annual regressions cluster standard errors by country. Country means use HC1 standard errors. FDI/GDP is in percentage points. A logarithm excludes zero and negative net flows, which are retained in the ratio comparison. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The negative slope across country means survives each individual country exclusion, ranging from -4.214 to -2.600 , but falls to -0.216 outside the original haven group. In the annual data with lagged risk observed, there are 2,397 positive, seven zero, and 160 negative net flows. The logarithm therefore excludes negative flows as well as zeros. Adding country effects to the dynamic log regression reduces its magnitude and precision. An annual FDI/GDP regression with country and year effects gives an imprecise positive coefficient of 0.250 (SE 1.412). Persistence, sample composition, and the outcome definition all matter. These estimates are retained as a check on the original evidence and are not used to establish an invariant investment response.

For the original cross-section of estimated inward profit misalignment relative to GDP, I estimate

$$PS_h = \alpha + \beta q_h^I + \delta_\tau \tau_h^{\text{stat}} + \delta' X_h + u_h, \quad (57)$$

where q^I is aggregate Investment Profile risk and X retains the original controls for capital flows, GDP growth, and the manufacturing share. On the same 116-country sample, adding the statutory rate changes the risk coefficient from -0.984 (SE 0.345) to -0.950 (SE 0.358). Excluding the original havens gives -0.155 (SE 0.195) in 99 countries. Tax controls alone do not remove the full-sample association, but its magnitude depends on the composition of hosts. These historical specifications use HC1 standard errors and omit the additional income, population, and controls for resource rents in Table 15. Their larger t statistics should not be read as a separate replication of the fully controlled estimates. The geographic comparison of profits and employment reported below addresses a different question, the separation of the reported and productive bases.

C.5 OECD Sources and Reporting Coverage

Fact 3 in Section 2.4 compares reported profits and employment directly. Their geographic shares must be calculated from identical cells for differences in coverage not to generate the comparison. The archived OECD country-by-country reporting file provides PROFIT, EMPLOYEES, and ASSETS, identified by reporting jurisdiction, counterpart jurisdiction, year, and profit grouping. Monetary quantities are in US dollars. PANELAI denotes the population with positive profits and _T all profit statuses. The code checks for duplicate observations before reshaping the file. It matches statutory rates by host and year, retains only individual foreign jurisdictions, and leaves missing observations unfilled. The public file of statutory tax rates includes national and subnational corporate income taxes. Country identifiers are matched using the archived World Bank country metadata.

The main comparison requires observed profits, nonnegative employment, and a statutory rate. It does not restrict the size of the effective tax rate or condition on a governance score, income, or observed assets. The balanced panel of profitable groups covers 141 hosts and 25 reporting parents. The balanced panel including all profit statuses covers 176 hosts and 27 parents. They contain 5,040 and 7,435 observations, respectively, including 45 and 87 cells with zero employment. Negative profits remain in the population including all profit statuses, while its aggregate profit denominator is positive in every reported comparison.

A balanced pair appears in every year from 2017 to 2021 after applying the observation requirements. The unbalanced comparison retains all available pairs in each year. The common-pair comparison intersects the balanced pairs across the two populations. These operations change the set of geographic cells, not the definition of either reported quantity. They do not hold the underlying firms fixed. Reporting coverage and the multinational-group threshold also limit the scope of the descriptive fact to the observed reporting population.

For a statutory threshold c , I calculate the low-tax share of quantity X as

$$S_t^X(c) = \frac{\sum_{(j,h) \in \mathcal{S}_t} \mathbf{1}\{\tau_{ht}^{\text{stat}} < c\} X_{jht}}{\sum_{(j,h) \in \mathcal{S}_t} X_{jht}}, \quad X \in \{\Pi, L\}, \quad (58)$$

where j denotes the reporting parent, h the host, Π reported profit, and L employment. The set $\mathcal{S}_t$ contains the same cells for both quantities within each reporting population. Rates and cutoffs are in percentage points. The reported difference is $S_t^\Pi(c) - S_t^L(c)$.

C.6 Tax Cutoffs and Reporting Composition

Section 2.4 reports the 2021 shares to distinguish the reported tax base from productive activity. Table 3 in the main text gives the full five-year comparison on fixed reporting pairs. The sample of profitable groups contains 1,008 pairs and the sample including all profit statuses 1,487 pairs. Profit shares exceed employment shares in each year. This ordering is descriptive and does not estimate the effect of reputation.

Table 17 reports the ranges of the profit share minus the employment share at every cutoff used in the analysis. The first column pools the five years, both populations, and balanced, unbalanced, and common-pair samples. The next two columns omit each individual host

or reporting parent from the balanced samples and recompute both denominators. Every difference in these comparisons is positive. The ranges describe variation across accounting comparisons, not confidence intervals from an assumed random sample.

TABLE 17: Profit share minus employment share across reporting comparisons

Cutoff (%)	Sample comparisons	Exclude one host	Exclude one parent
10	2.44–12.55	0.42–15.58	0.44–14.86
15	6.67–20.48	2.92–25.51	1.17–22.80
20	14.75–30.78	11.86–41.33	10.05–44.21
25	11.78–44.11	10.61–52.12	5.84–55.58

Notes: Minimum and maximum differences in percentage points, pooling both reporting populations and all five years. Sample comparisons include balanced, unbalanced, and common balanced pairs. Each exclusion recomputes shares on the remaining cells of a balanced sample.

The earlier construction required both positive employment and positive tangible assets to compare all three quantities on identical cells. It contains 980 balanced pairs for profitable groups and 1,436 pairs for all profit statuses. The ordering of profit and employment shares holds in every balanced and unbalanced year and in all 6,240 exclusions of individual hosts across the four cutoffs. With common pairs, the sample including all profit statuses at the 10% cutoff has a reversal of 0.03 percentage points in 2017. Excluding the United States as a reporting parent from that year’s balanced sample including all profit statuses gives reversals of 0.80 and 0.16 points at 10% and 15%. No other omission of a reporting parent in that construction reverses the ordering. The 20% and 25% comparisons remain positive throughout both sample definitions. The main definition avoids requiring asset data for a comparison of profits and employment, while these narrower comparisons show the effect of that extra restriction.

The comparison of profits and assets is more sensitive and is not used as a separate empirical finding. Tangible assets omit intangible capital, while accounting profits can include distributions from other group entities. These measurement features further limit a structural interpretation of geographic shares. The replication files preserve every cutoff, population, year, and country omission, including the exceptions above.

C.7 Tax Burdens and Government Tax Announcements

The relationships in Sections 2.1 and 2.4 do not directly record whether a government honors a dated tax promise. This distinction is essential when interpreting the government’s probability of honoring its announcement in Section 4.2. I first show why an accounting tax gap is not an observed policy revision, then separate the probability of a commitment type from the probability that a tax promise is honored. With current accrued income tax written as tax on the applicable base plus the remaining adjustments,

$$\frac{\text{Accrued tax}}{\text{Accounting profit}} - \tau = \frac{\tau(\text{Taxable profit} - \text{Accounting profit}) + \text{Tax adjustments}}{\text{Accounting profit}}. \quad (59)$$

Loss offsets, credits, depreciation, and sectoral rules can change this gap even when the government implements the announced rules. Fees outside income tax do not enter its numerator. Equation (59) concerns current accrued income tax. Cash payments and deferred tax expenses

are different accounting quantities. Taxes and profits must refer to identical reporting cells before a ratio is constructed. In particular, a denominator of adjusted profits available for only some parents cannot be paired with taxes summed over all parents. In the model, the implemented rate multiplies a defined tax base after shifting. Dividing tax payments by that same base recovers the rate. Replacing the denominator with accounting profits requires a measurement model that is not imposed here.

A tax revision must likewise be measured relative to a dated announcement applying to the same investment. A historical liability on a completed transaction cannot be avoided by reallocating current profits. Anticipated reforms and sectoral changes are not automatically violations of earlier promises. These distinctions explain why accounting tax gaps and broad measures of country risk cannot supply direct observations of a tax revision or investors' initial probability of a commitment type in the model.

Let p be the probability investors initially assign to a commitment type, and let σ be the opportunistic type's probability of honoring its announcement. The probability that the observed government honors its announcement is $\pi = p + (1 - p)\sigma$. Even if π were known without measurement error, it would identify only

$$p \in [0, \pi], \quad \sigma = \frac{\pi - p}{1 - p} \quad \text{for } p < 1. \quad (60)$$

For any p in that interval, the implied σ lies in $[0, 1]$ and reproduces the same probability of honoring the announcement. Broad governance indicators cannot narrow the interval without additional restrictions relating their measurement to types and behavior. In the two-period model, by contrast, the government's tax strategy and Bayes' rule jointly determine how investors update that probability after observing the implemented tax.

A direct empirical test would need a dated policy promise, exposure determined before its revision, and a comparison separating the direct tax change from its effect on expectations about later promises. Selecting episodes from declines in the same risk score used as an outcome would not provide that comparison. The ICRG regressions document associations of contract risk with taxes, FDI, and reported profits with different degrees of precision and sensitivity. The geographic shares separately motivate the distinction between investment and reporting. Neither identifies the effect of breaking a dated tax promise. The model derives that comparison from specified beliefs and incentives.